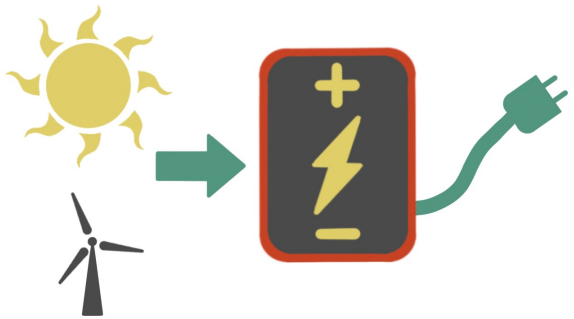
Cell behaviors and models



Ania's BMS Course

by Ania Mitros, PhD

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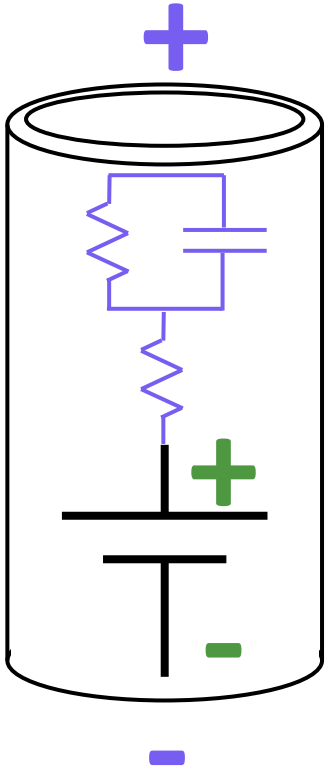
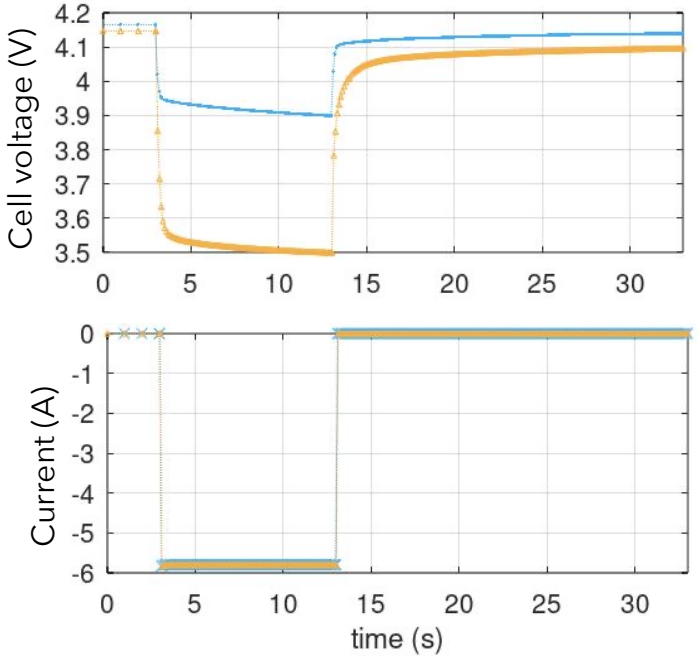
https://ofb.net/~ania/teaching/BMS-course/

Cell behaviors and models

To understand battery management, we need a basic understanding of some battery cell behaviors.

- Terminology
- OCV: Open Circuit Voltage
- Cell impedance
- Time-dependent responses
- Aging
- Another learning resource





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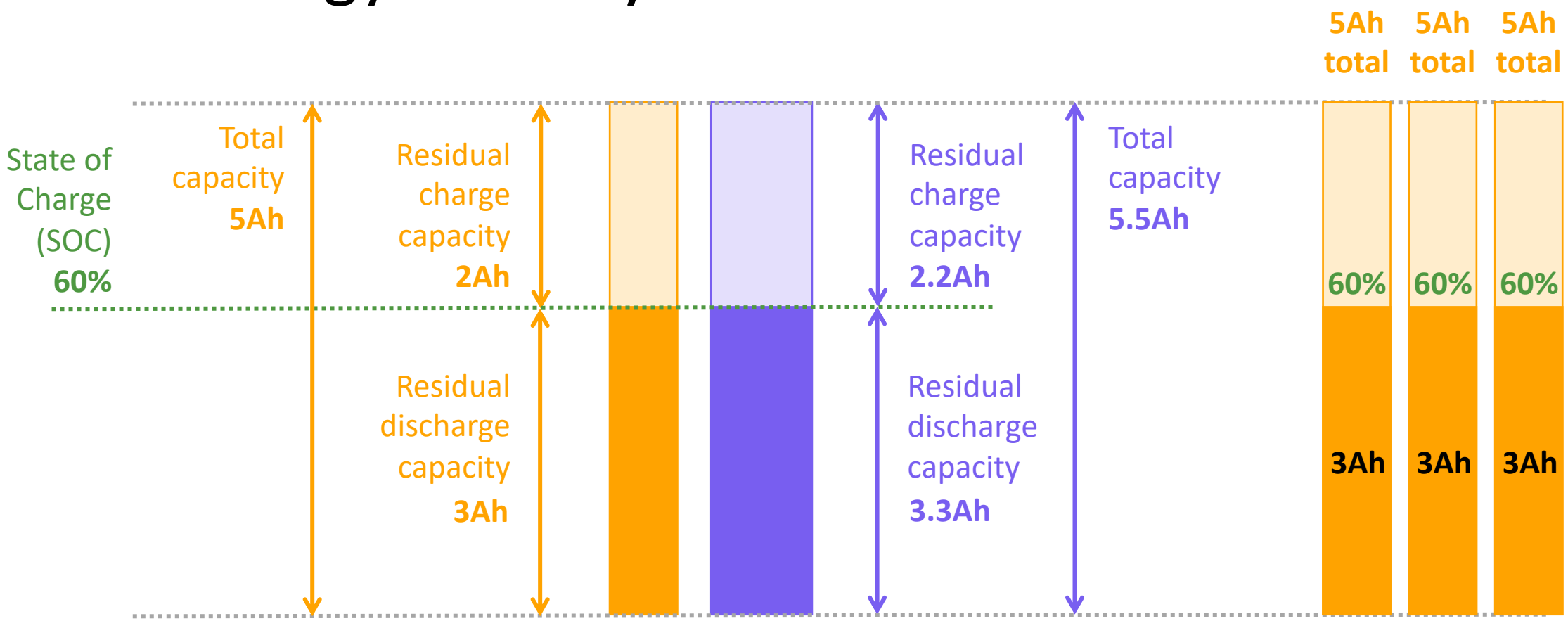
Terminology: Battery cell state



Capacity (Ah): Amount of charge in charge-measuring units like amp hours (Ah) or Coulombs (C)

SOC (%): What fraction of the battery is full in percent (%) $SOC = \frac{residual\ discharge\ capacity}{total\ capacity}$

Nominal capacity (Ah): Amount of charge a battery can supply to a load or can discharge when new (not aged)

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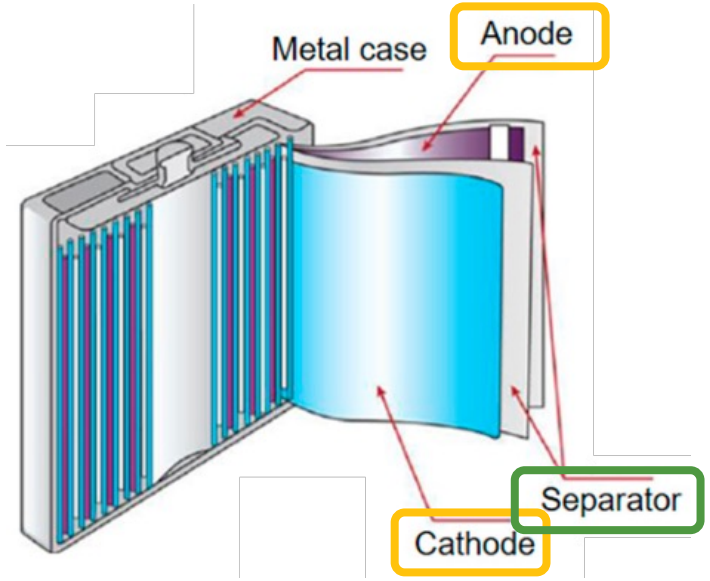
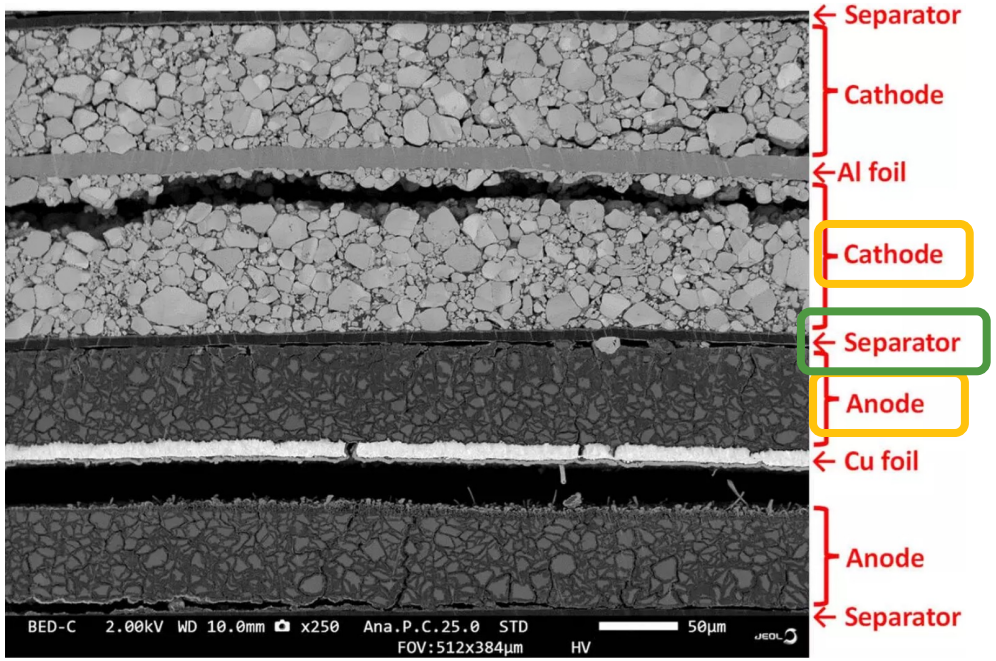
Battery cell construction

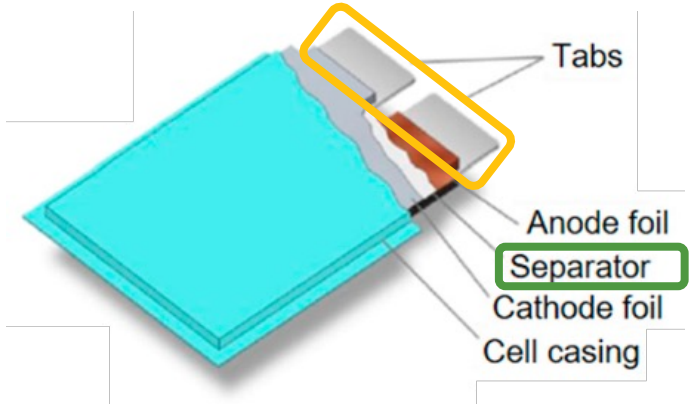
Prismatic cell

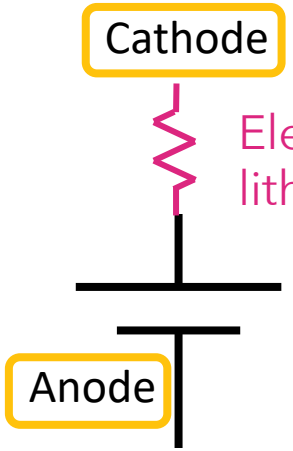


Cylindrical cell



Pouch cell





Electrolyte: medium through which lithium ions flow

Cylindrical cell image: DOI: 10.1149/ 1945-7111/ab7fb6

Pouch and prismatic drawings: https://doi.org/10.1016/j.est.2020.101985

SEM cross-sections: https://www.jeolusa.com/BLOG/pristine-cross-sections-of-batteries

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Cell behaviors and models

A basic understanding of some battery cell behaviors, enough to optimize battery management architecture.

- Terminology
- **OCV: Open Circuit Voltage**
- Cell impedance
- Time-dependent responses
- Aging
- Another learning resource

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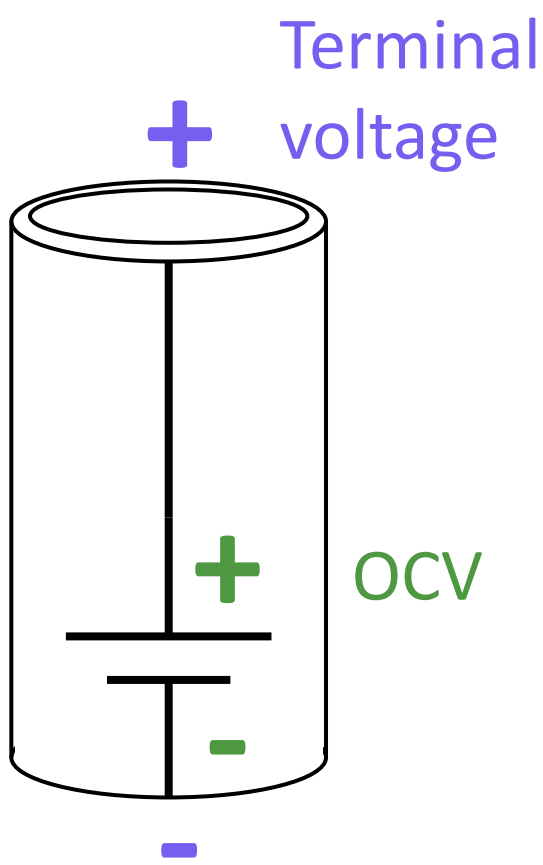
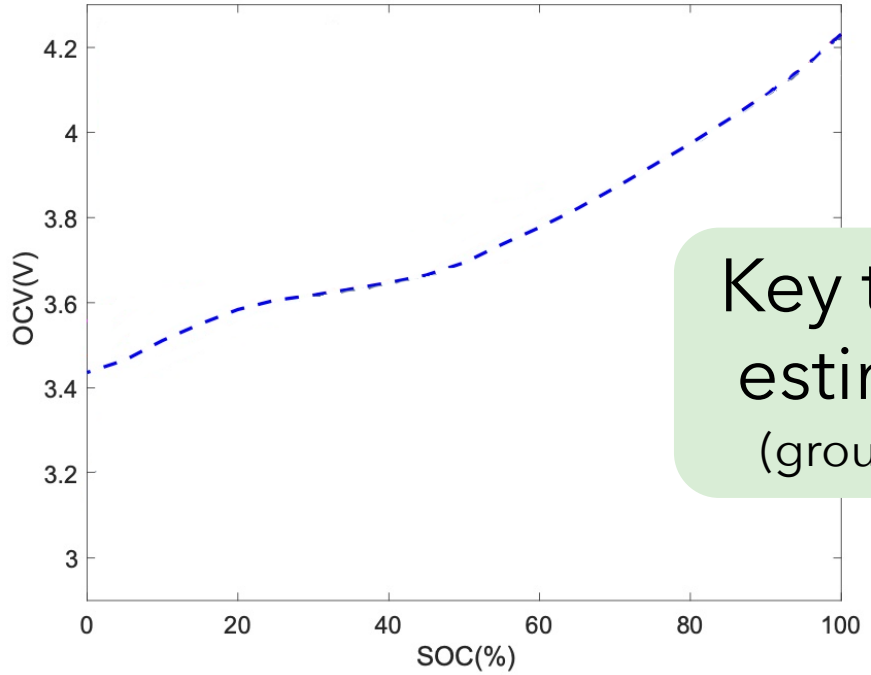
OCV: Open Circuit Voltage

OCV: Cell voltage when disconnected (open)



- OCV is a function of SOC (State of Charge)



Key to SOC estimation
(ground truth)

- OCV does not include I×R drop due to current flow nor time-dependent effects.

Curves from: Zhang 2018, doi:10.3390/en11092408

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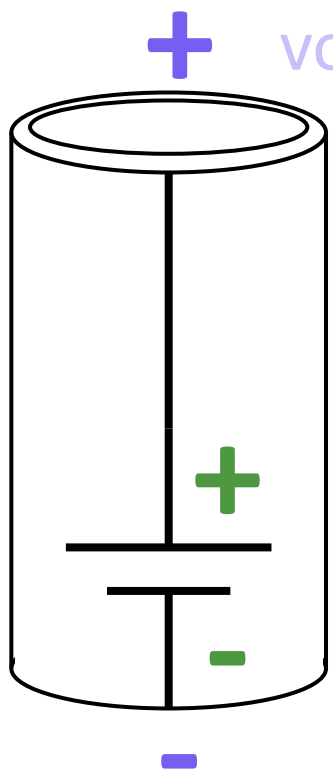
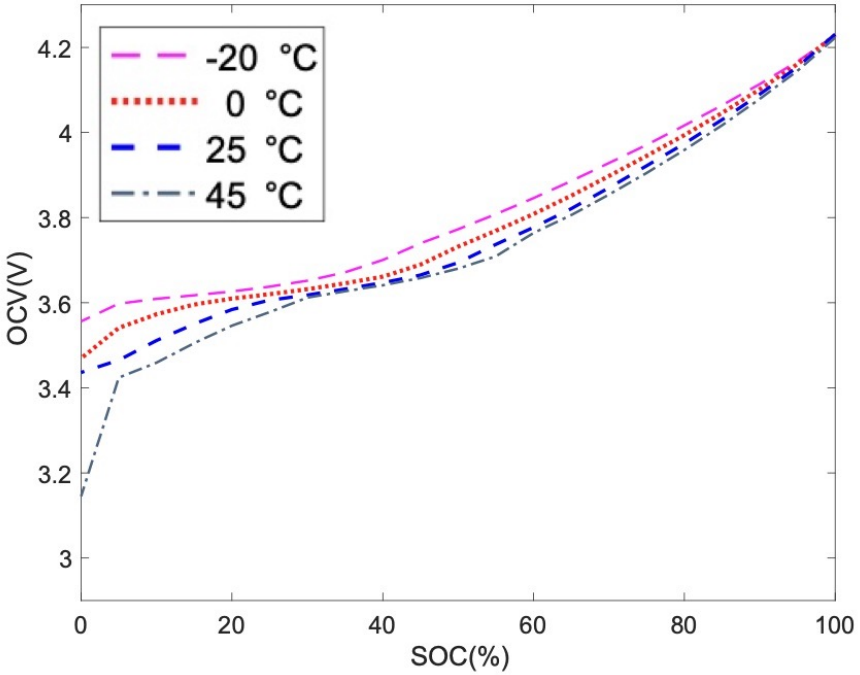
OCV: Open Circuit Voltage and temperature

OCV: Cell voltage when disconnected (open)



- OCV is a function of SOC (State of Charge)



- OCV is a function of temperature

Curves from: Zhang 2018, doi:10.3390/en11092408

https://ofb.net/~ania/teaching/BMS-course/

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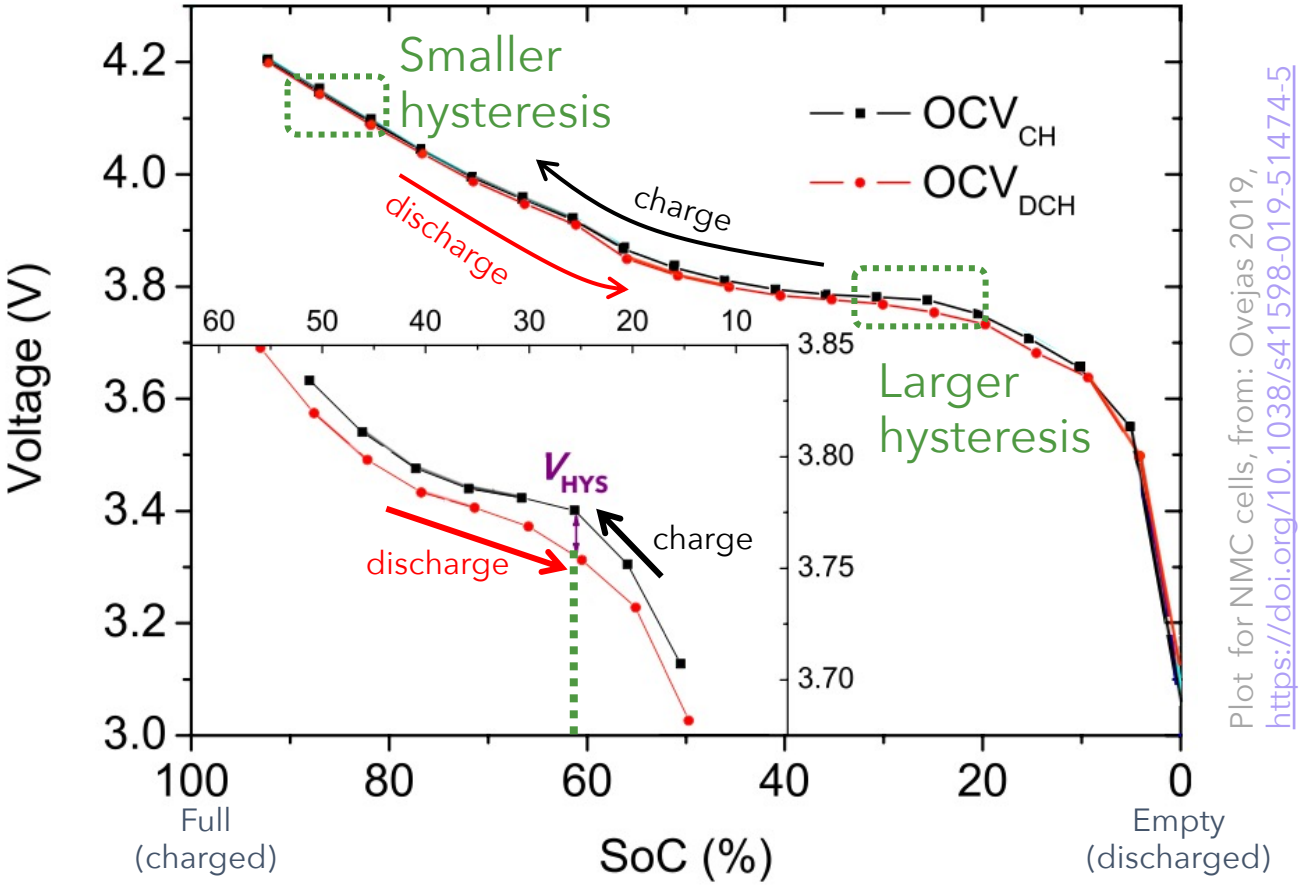
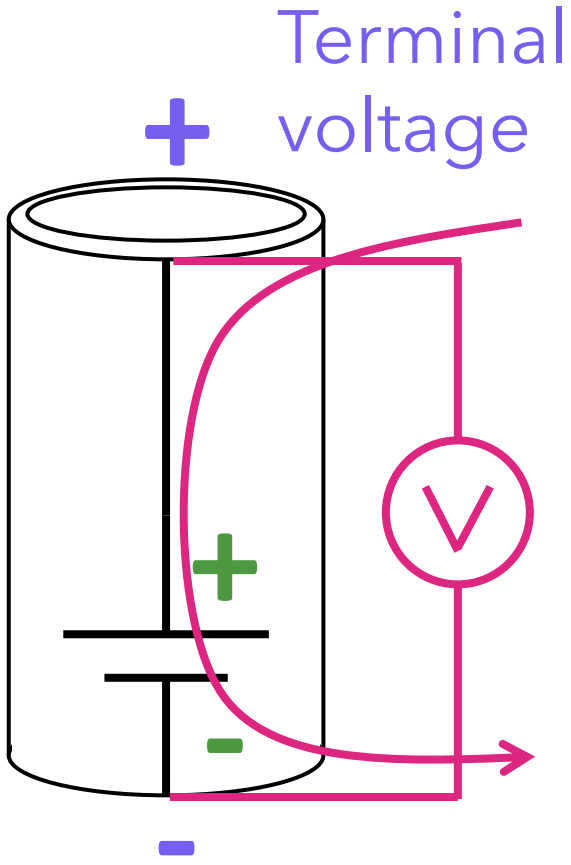
OCV and hysteresis

Hysteresis: difference between OCV curves in **charge** versus in **discharge**.



Plot for NMC cells, from: Ovejas 2019, https://doi.org/10.1038/s41598-019-51474-5



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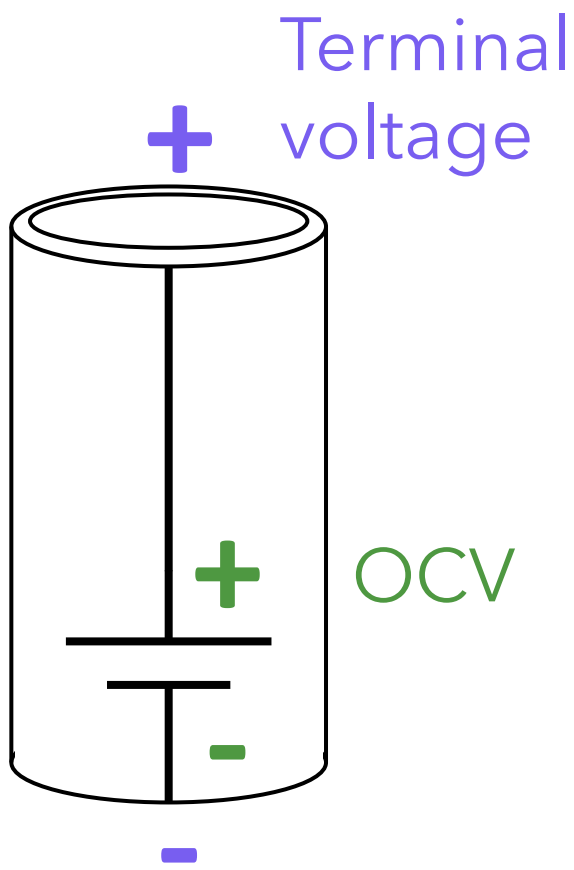
OCV: Key points

OCV: Cell voltage when disconnected (open)



- OCV is a function of SOC (State of Charge)
- OCV is a function of temperature
- OCV is subject to hysteresis ← Smallest effect of the three

OCV is key to SOC estimation
(ground truth)

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Cell behaviors and models

A basic understanding of some battery cell behaviors, enough to optimize battery management architecture.

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- **Cell impedance**
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- Aging
- Another learning resource

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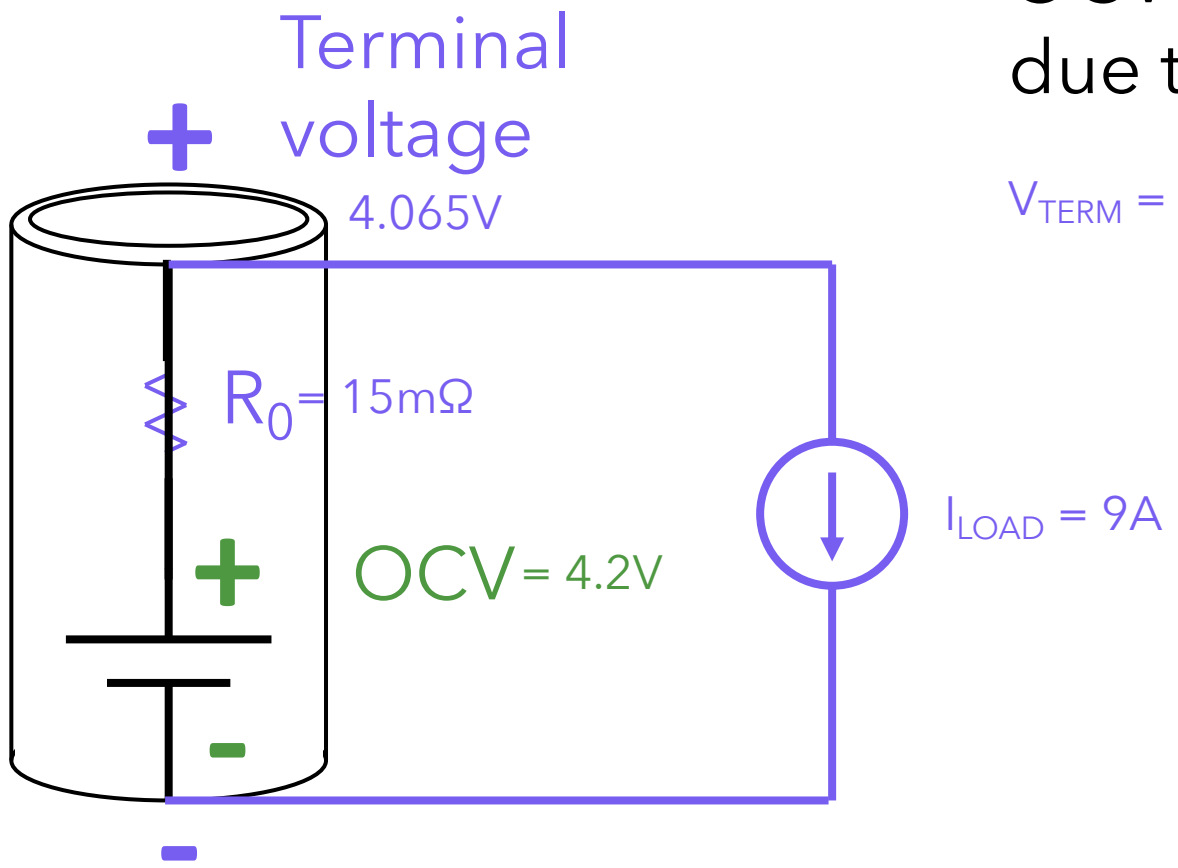
Cell impedance



OCV does not include I×R drop due to current flow.

V_{TERM} = OCV - (9A)×(15mΩ) = 4.2V - 0.135V = 4.065V

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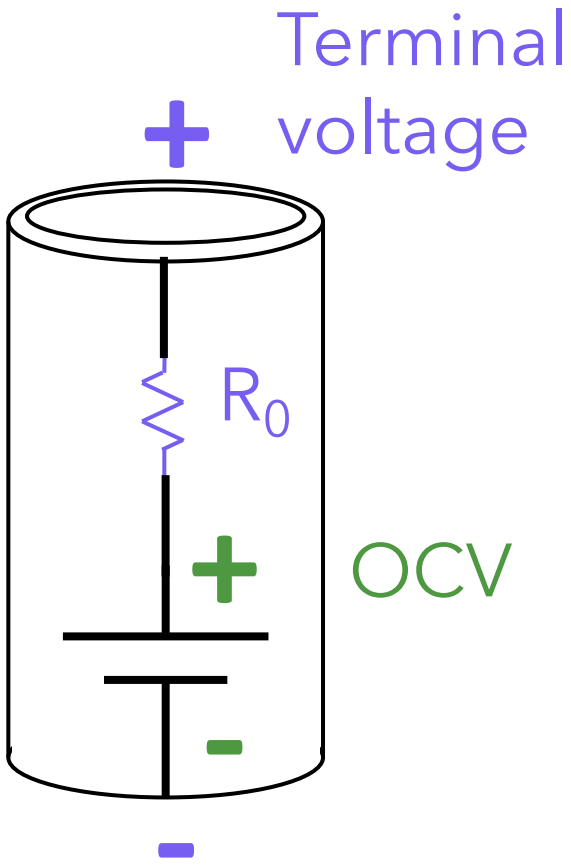
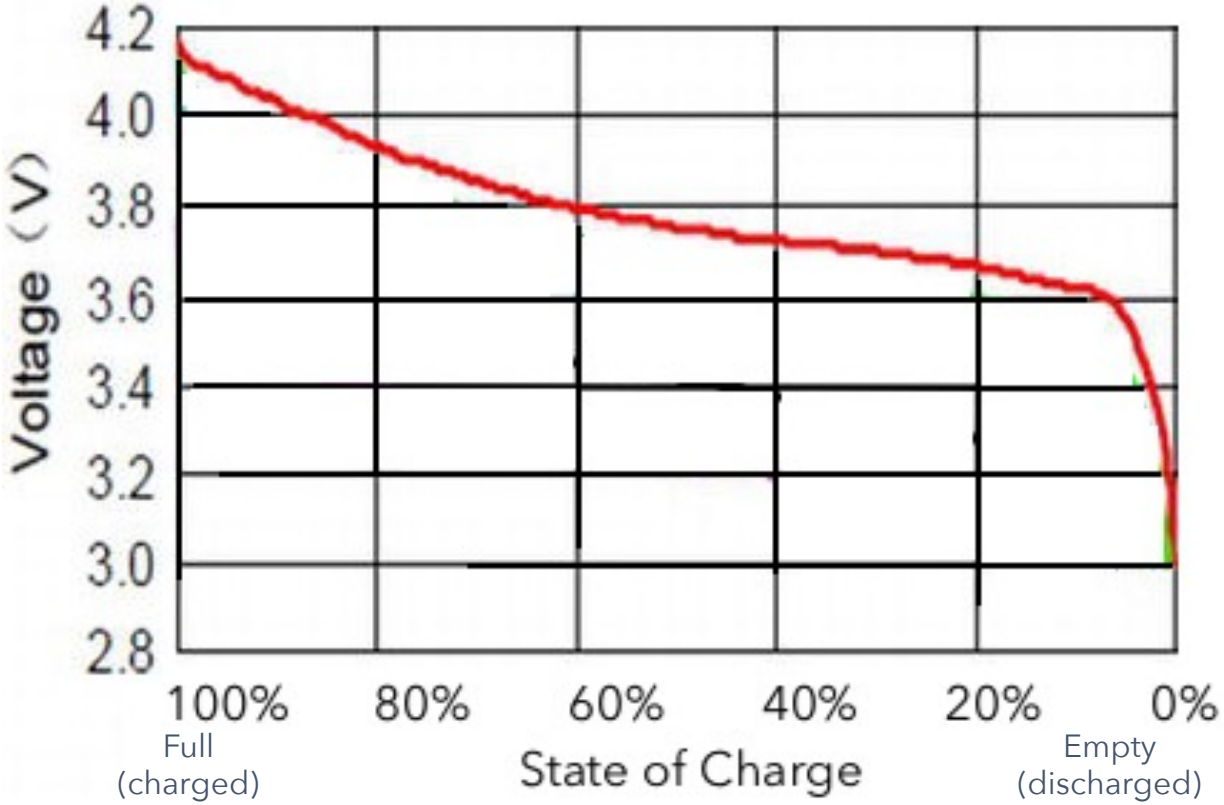
Cell impedance





Cell curves: © 2021 The Author(s). Licensee IntechOpen, distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/ by/3.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. http://dx.doi.org/10.5772/intechopen.99851

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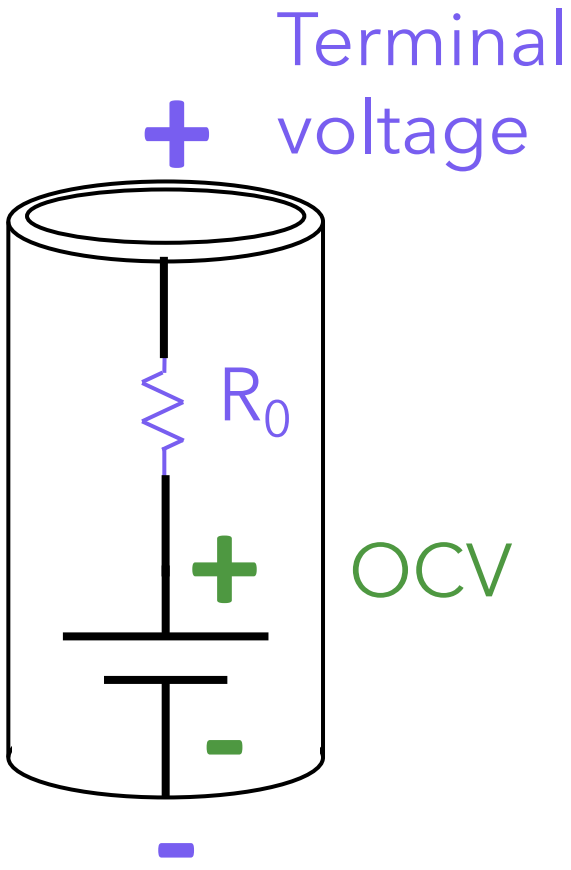
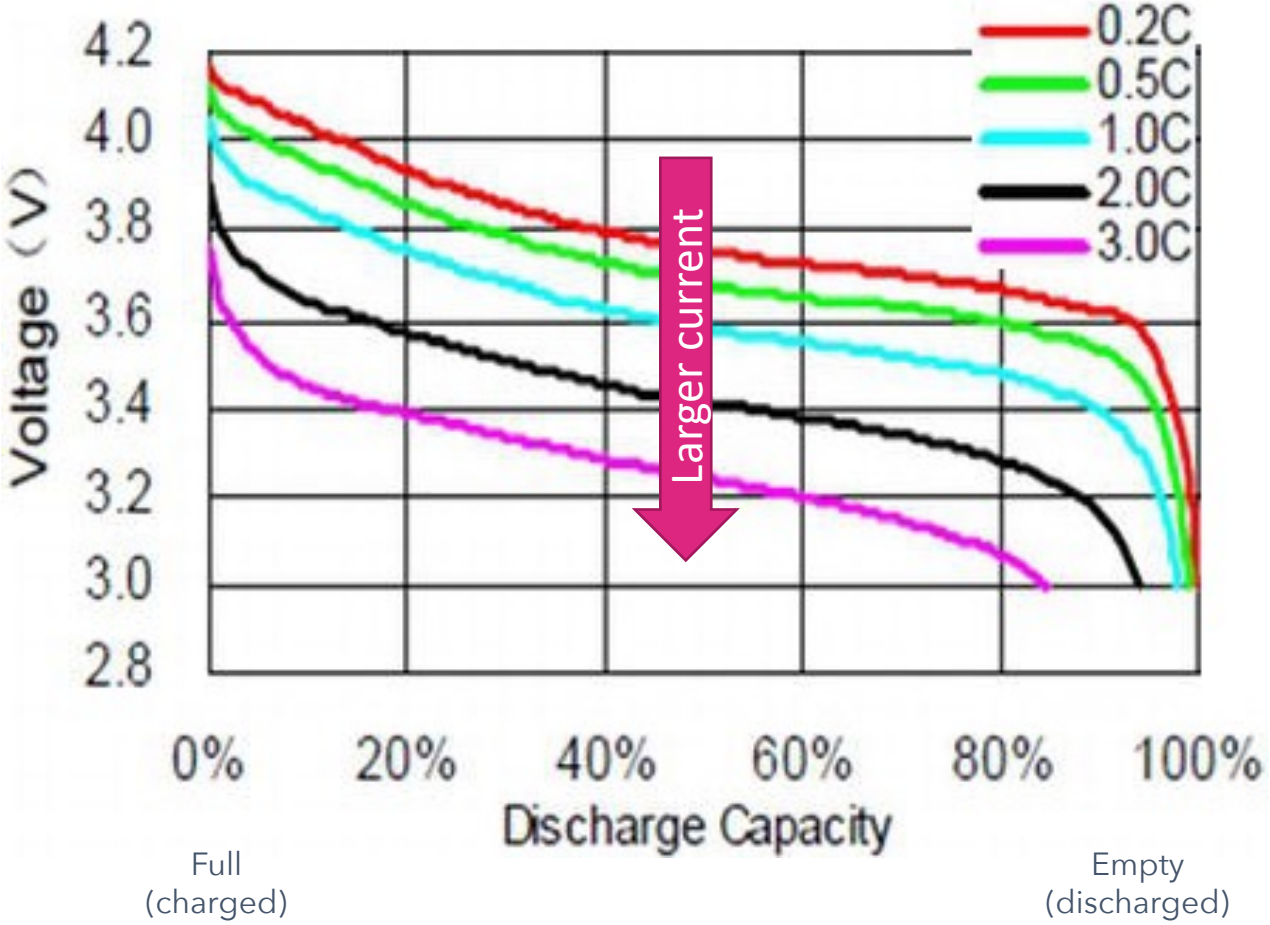
Cell impedance

R0 models electrolyte resistance and contact resistance





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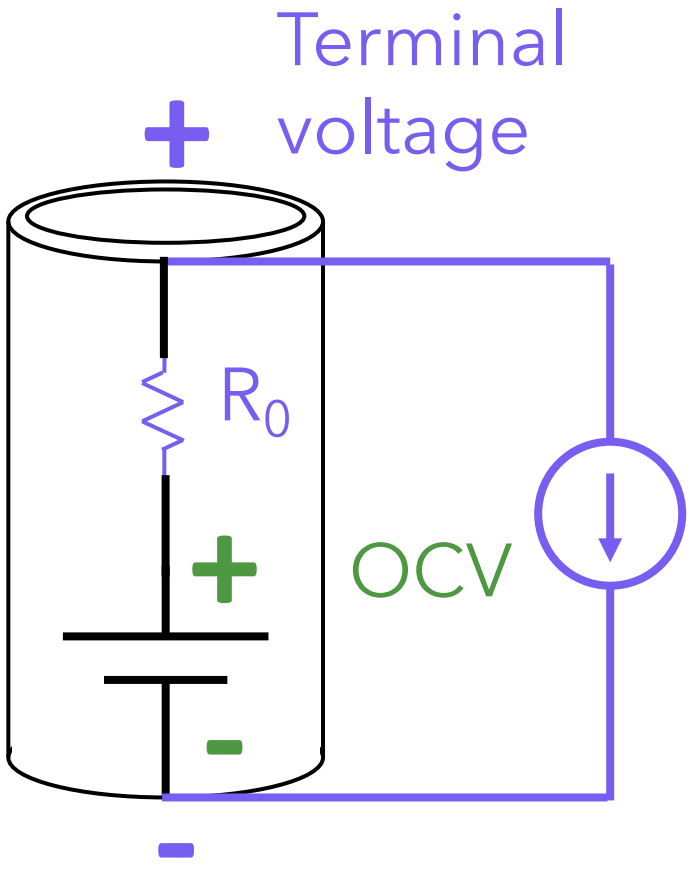
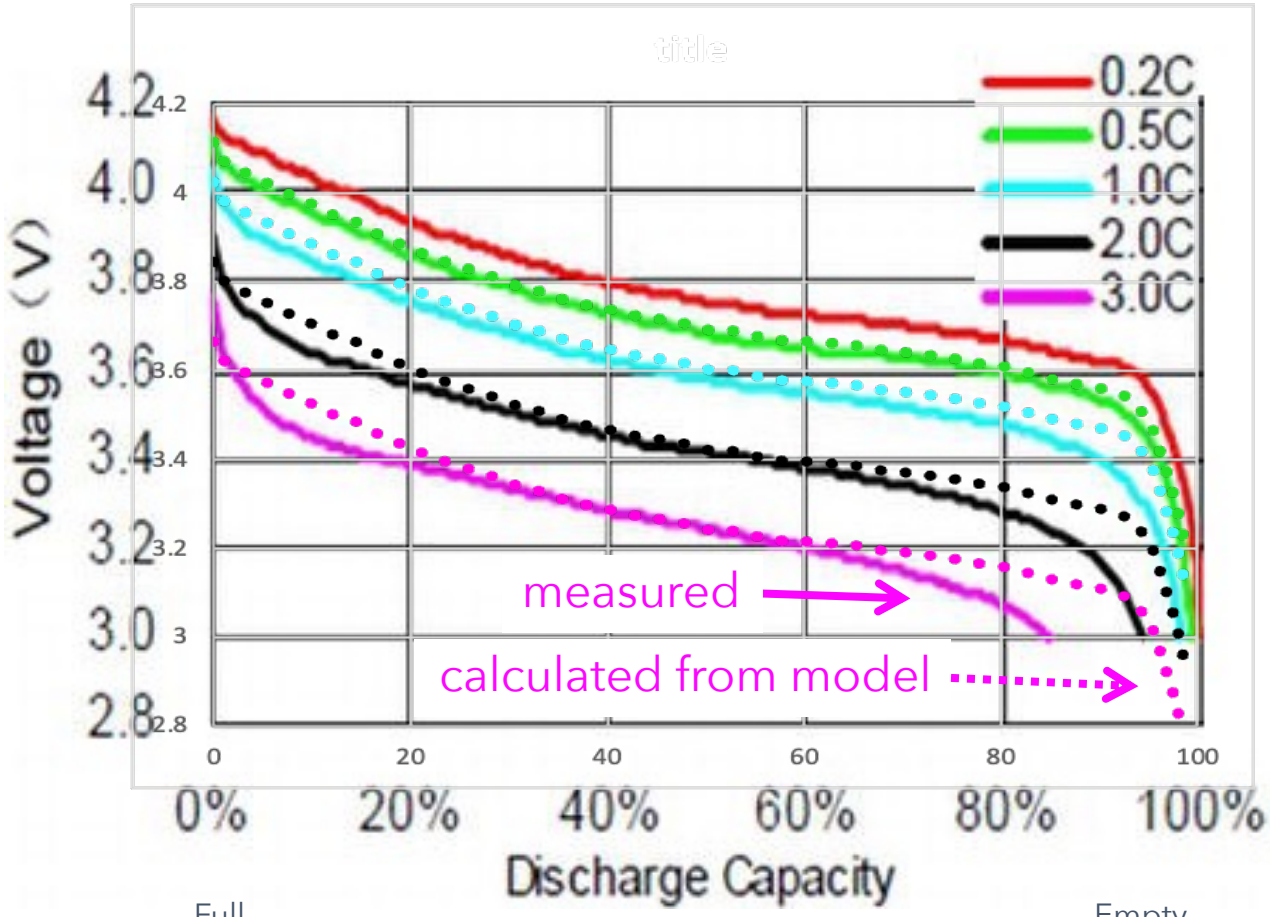
Cell impedance





Full (charged)

Empty (discharged)

Cell curves: © 2021 The Author(s). Licensee IntechOpen, distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/ by/3.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. http://dx.doi.org/10.5772/intechopen.99851

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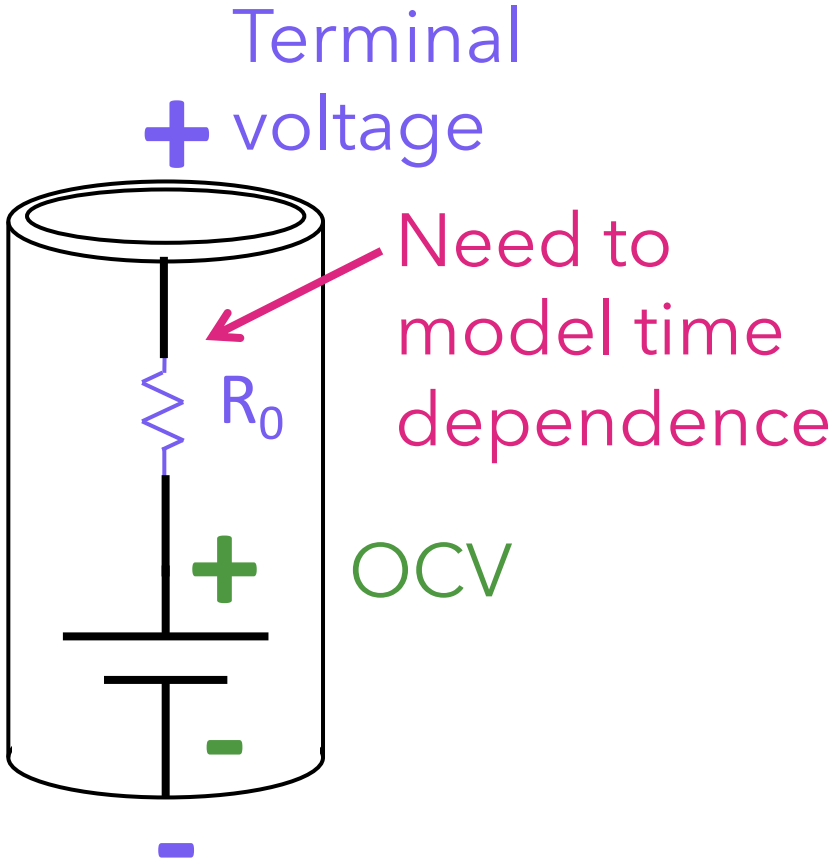
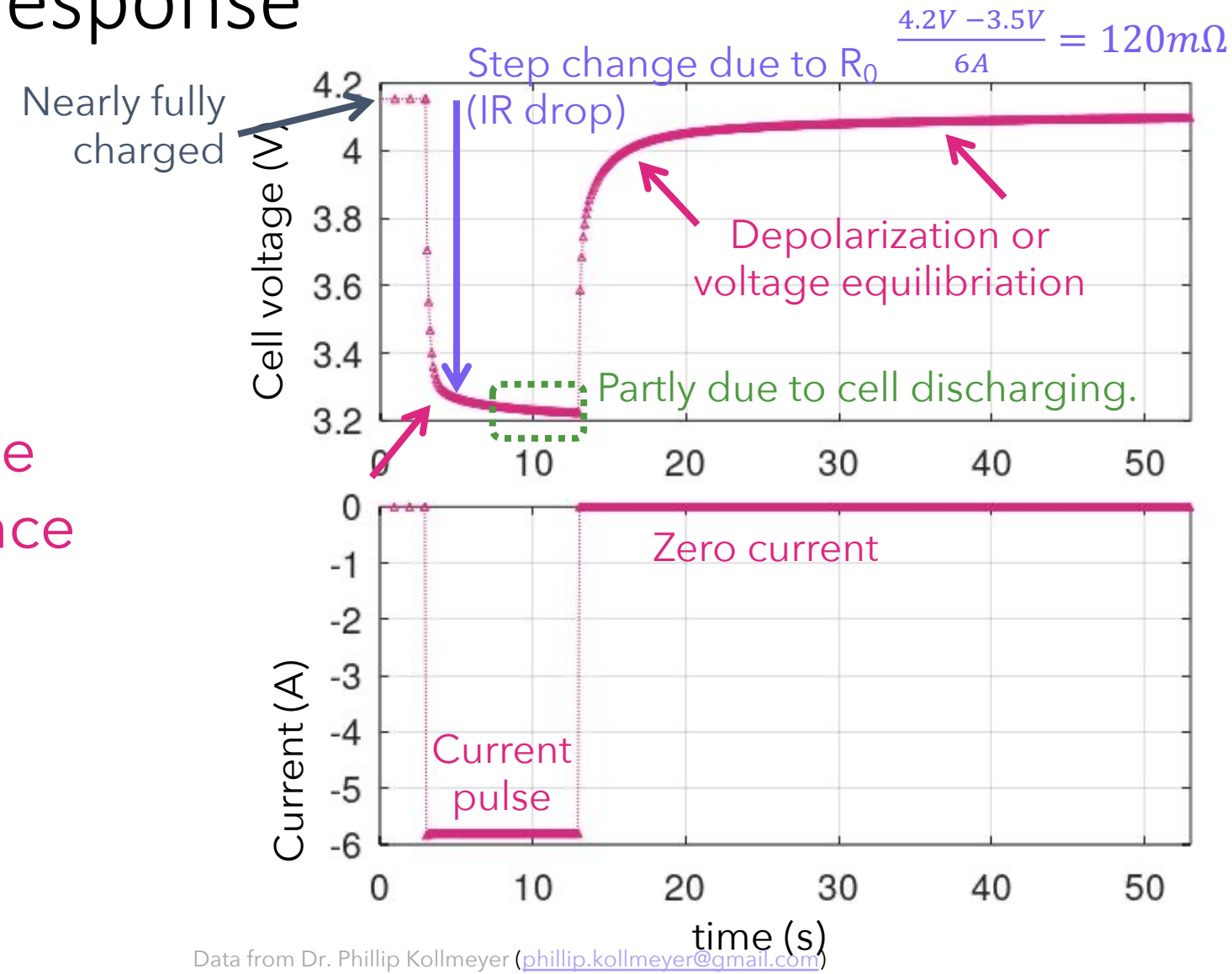
Time-dependent response





Data from Dr. Phillip Kollmeyer (phillip.kollmeyer@gmail.com)
https://data.mendeley.com/datasets/wykht8y7tg/1#file-57a2002f-d618-4dd8-96e3-37ffb6bc26c5

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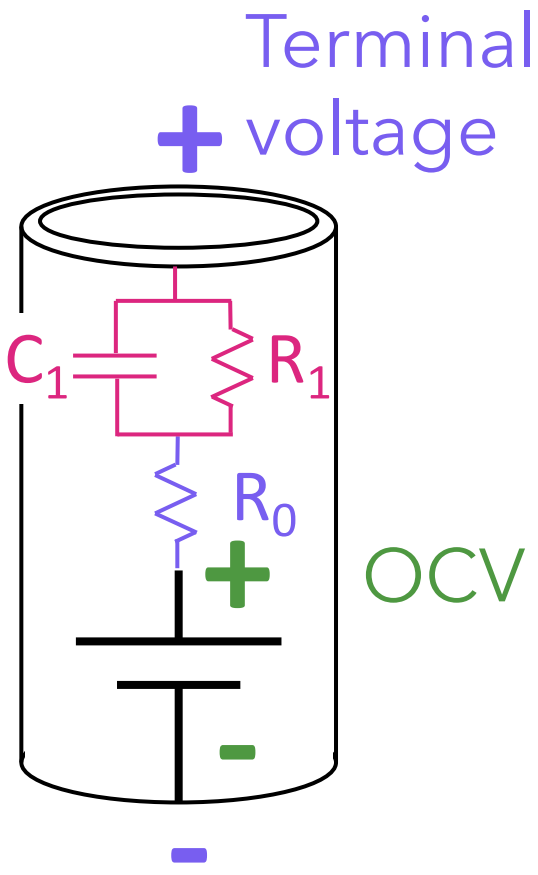
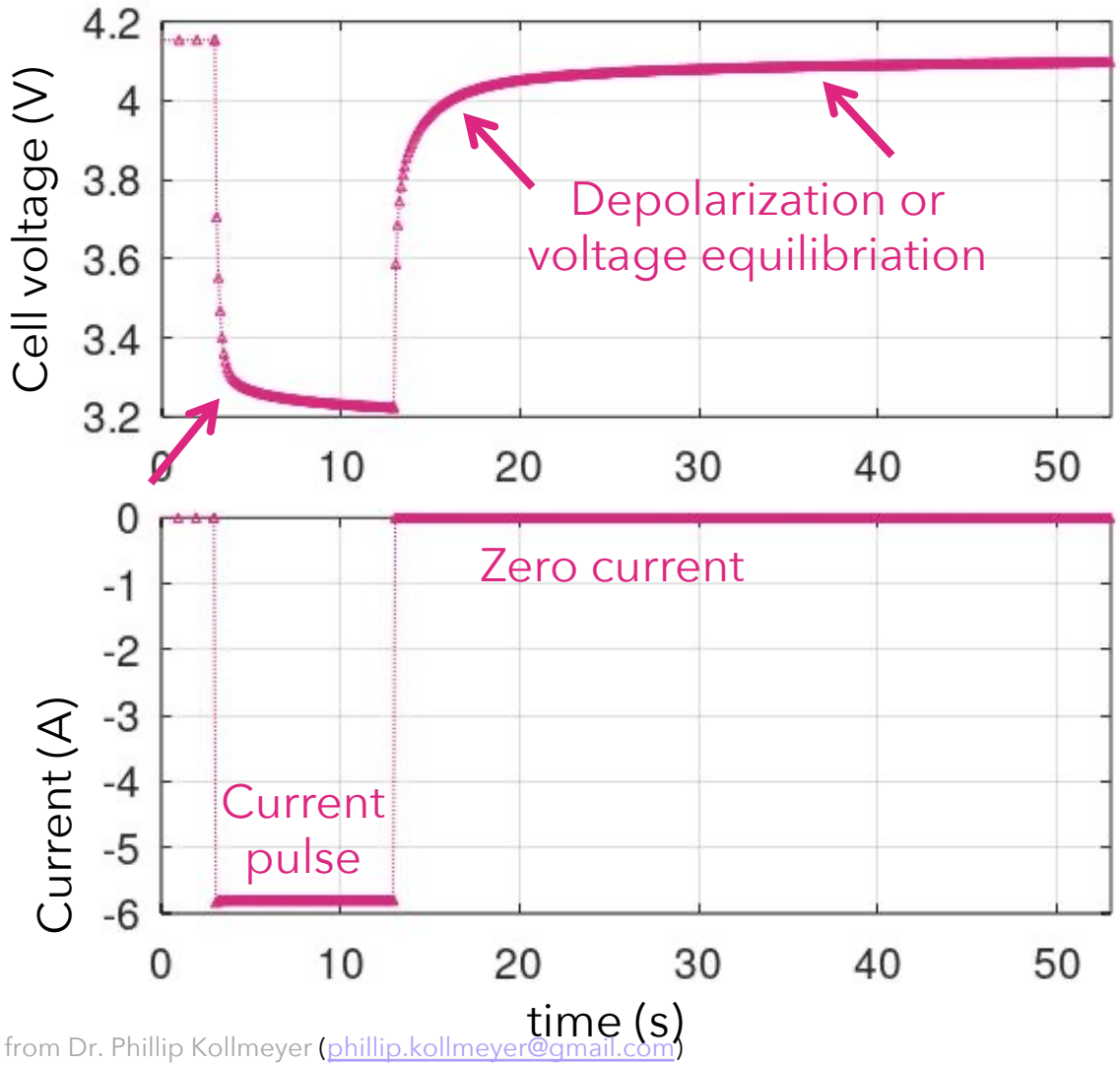
Time-dependent response





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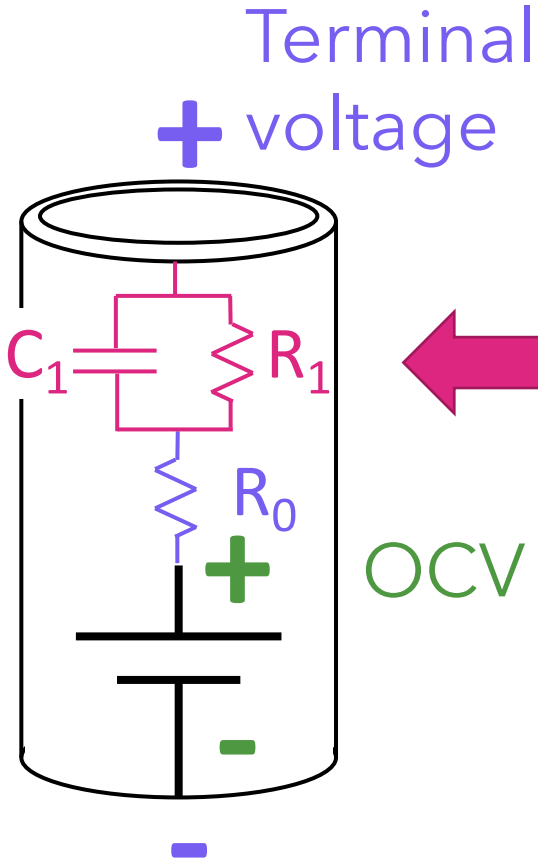
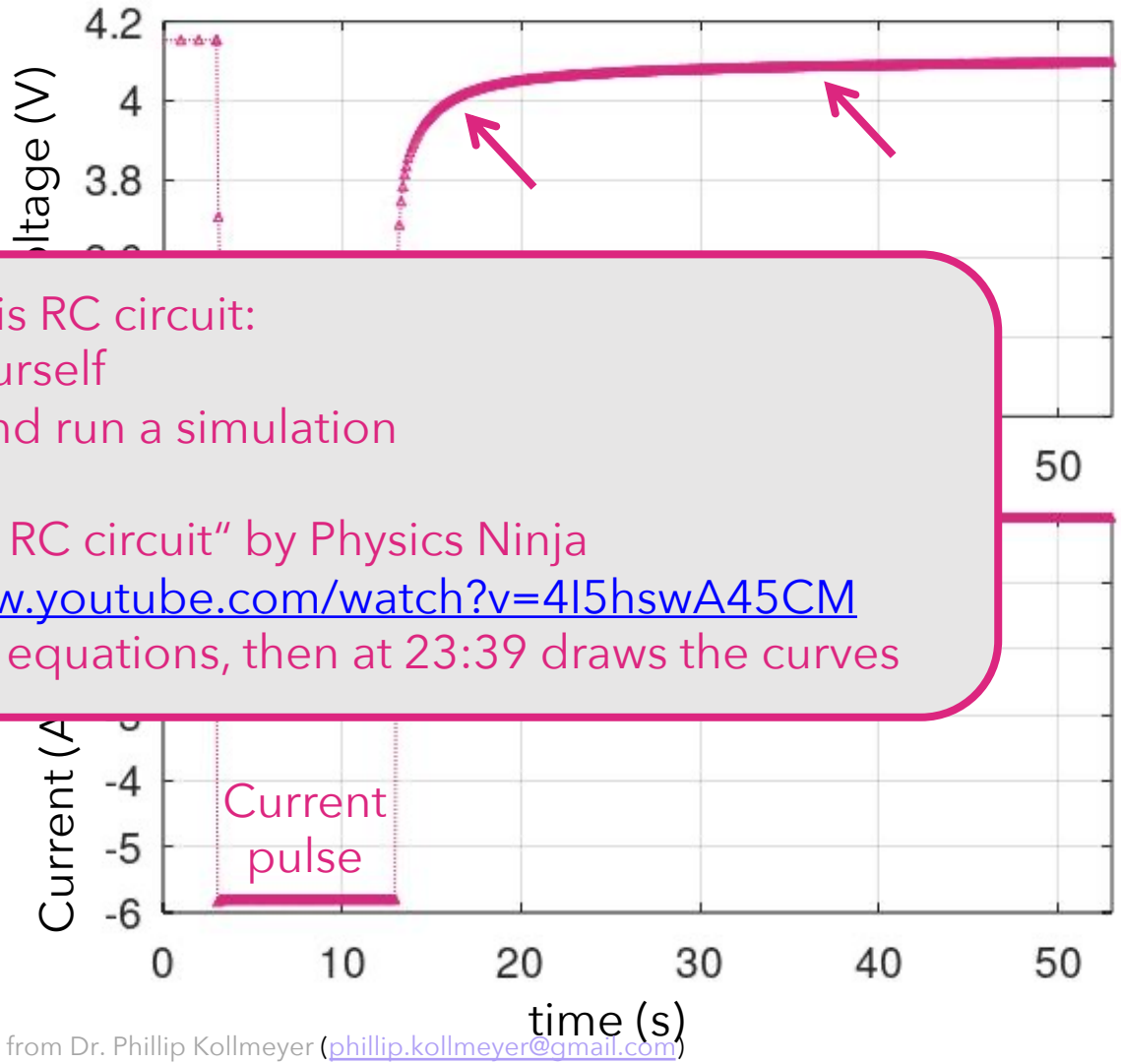
Time-dependent response



For refresher on this RC circuit:

- Do the math yourself
- Install LTspice and run a simulation
- Watch a tutorial
  - Ex. "Parallel RC circuit" by Physics Ninja
  - https://www.youtube.com/watch?v=4I5hswA45CM
  - Derives the equations, then at 23:39 draws the curves



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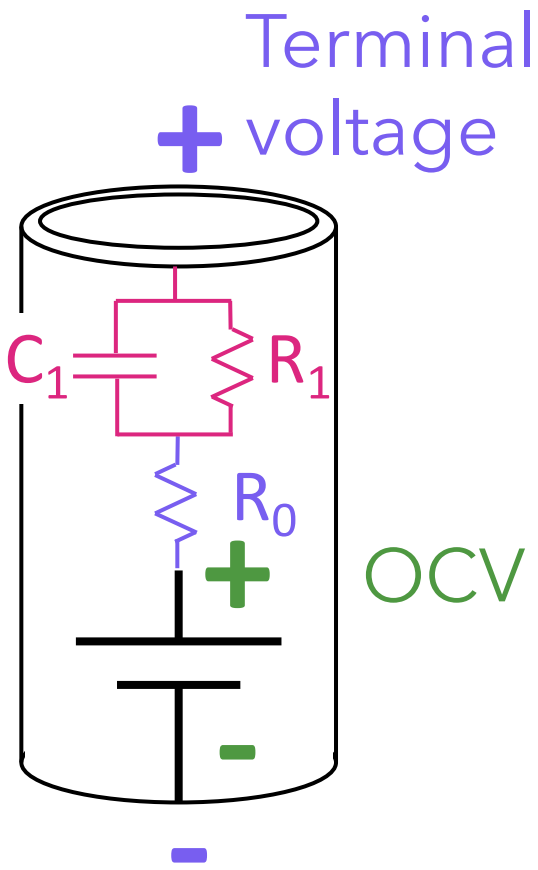
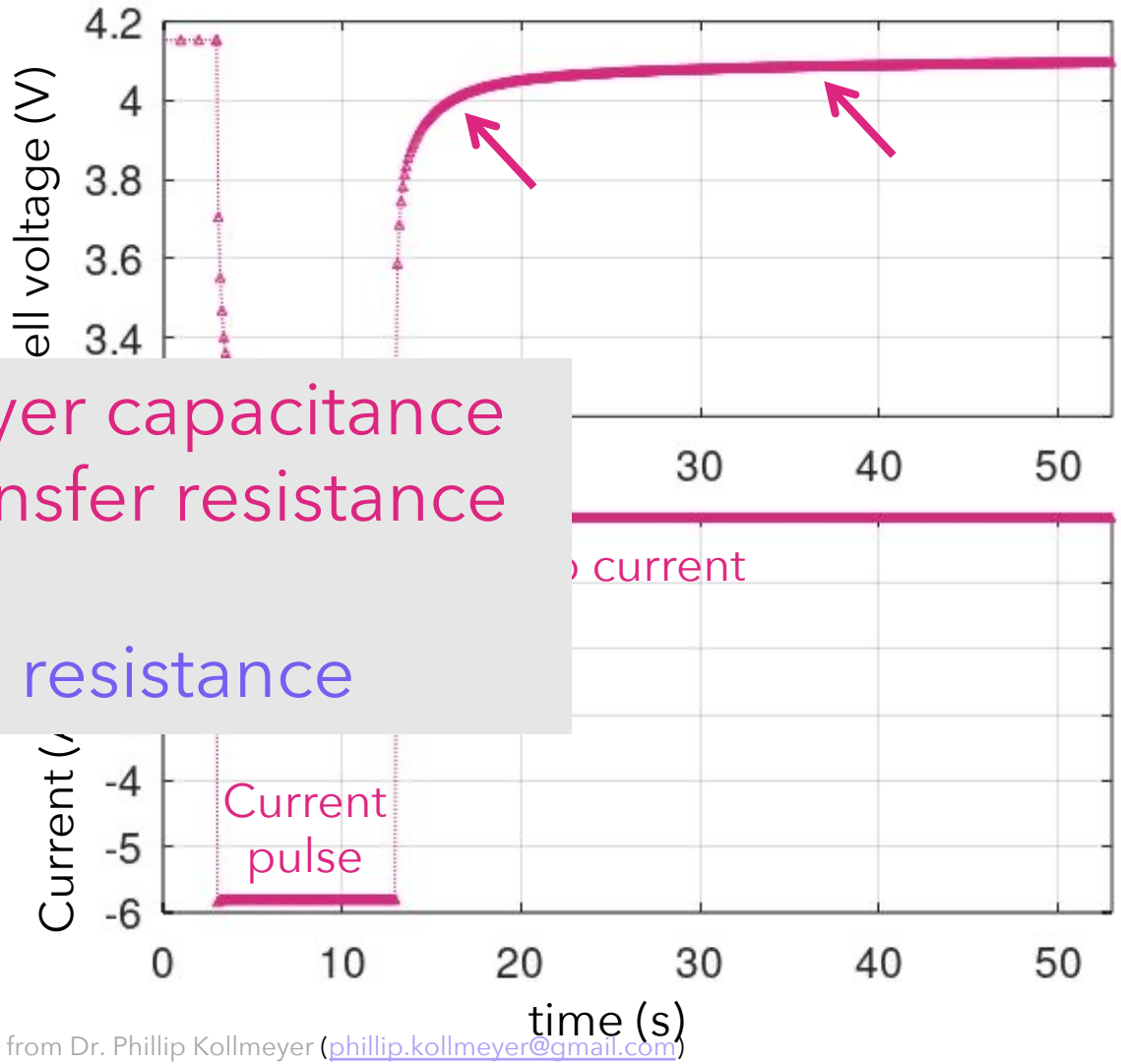
Time-dependent response



C1: double-layer capacitance
R1: charge transfer resistance

R0: electrolyte resistance



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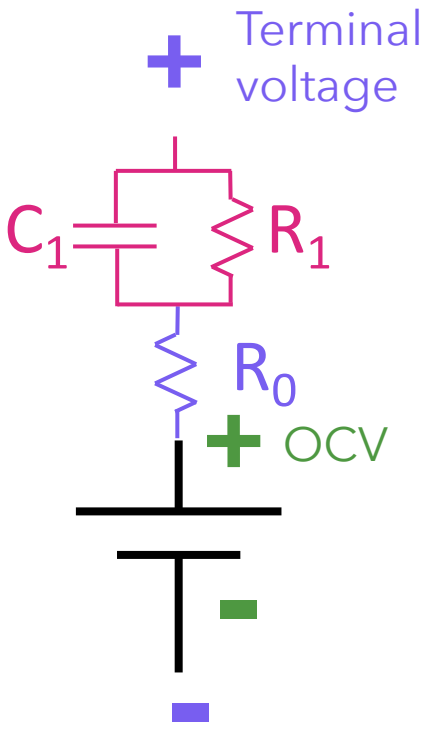
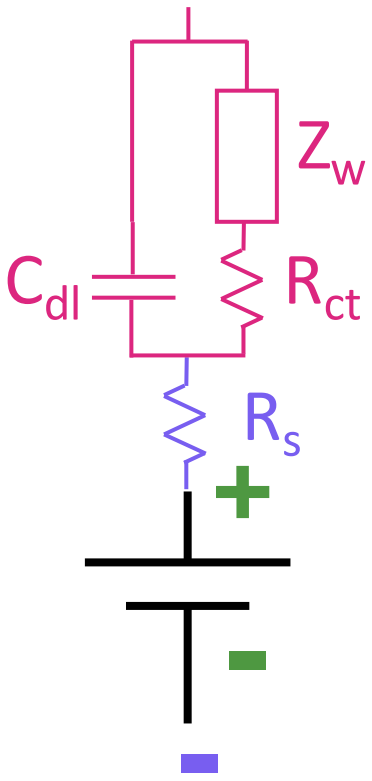
Family of models

Randles circuit: based on electrochemical principles… maybe





Warburg impedance

$$Z_w = \frac{A_w}{\sqrt{j\omega}}$$

- R_s models the **electrolyte resistance**
- R_{ct} is the **charge-transfer resistance** that models the voltage drop at the interface of the electrolyte and the electrode particles due to a load
- C_{dl} is the **double-layer capacitance** that models the effect of charge build-up at the interface of the electrolyte and the electrode particles
- Z_W is the **Warburg impedance** that models the diffusion of lithium ions in the electrodes.

No simple differential-equation representation of a Warburg impedance → Hard to implement in code

Go deeper: ECE5710-Notes02.pdf from Prof. Gregory Plett
http://mocha-java.uccs.edu/ECE5710/ECE5710-Notes02.pdf

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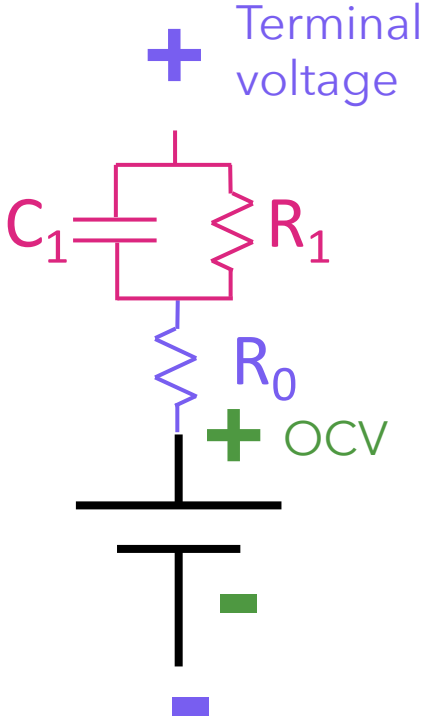
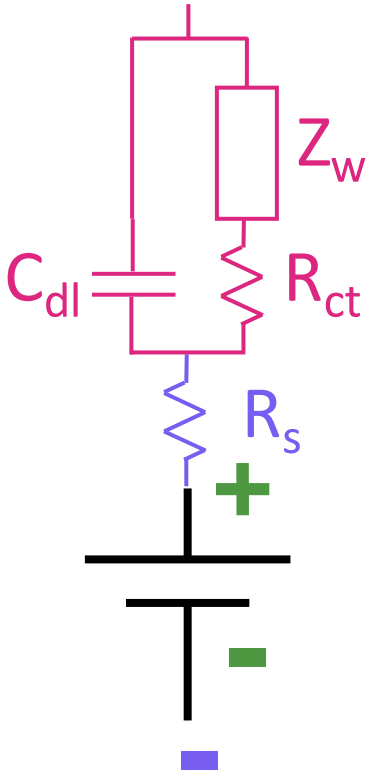
Family of models



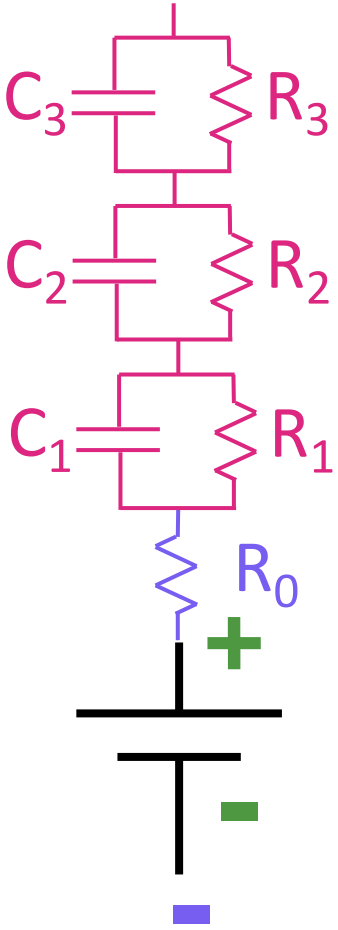
Randles circuit: based on electrochemical principles



Warburg impedance

$$Z_w = \frac{A_w}{\sqrt{j\omega}}$$

1 or 2 often suffice



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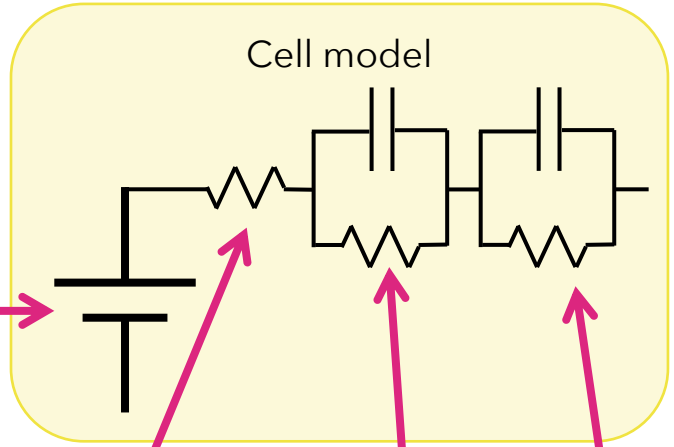
Aging and State of Health (SOH)



Total capacity decreases with age

Electrolyte and contact impedances increase as cell ages

SEI (Solid Electrolyte Interphase) grows with age

Charge transfer resistance increases with age

Aging increases with: calendar time, high temperatures, high currents, high SOC.

Aging affects: Capacity & impedance & susceptibility to failure

State of Health (SOH): Somewhat ambiguous. Can refer to capacity fade and/or impedance increase.

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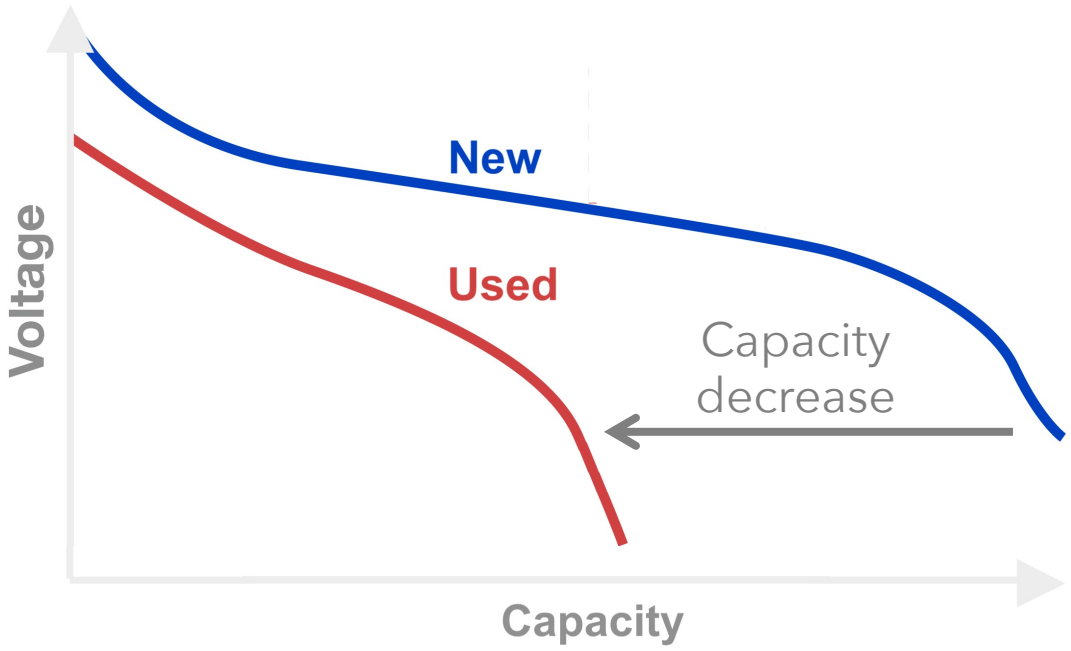
Aging: Capacity

State of Health (SOH) is often quantified as:

$$SOH = \frac{Q_{available}}{Q_{nominal}}$$

(This ignores degradation of cell resistance with age)



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Capacity loss: Time, SoC, Temperature, Chemistry

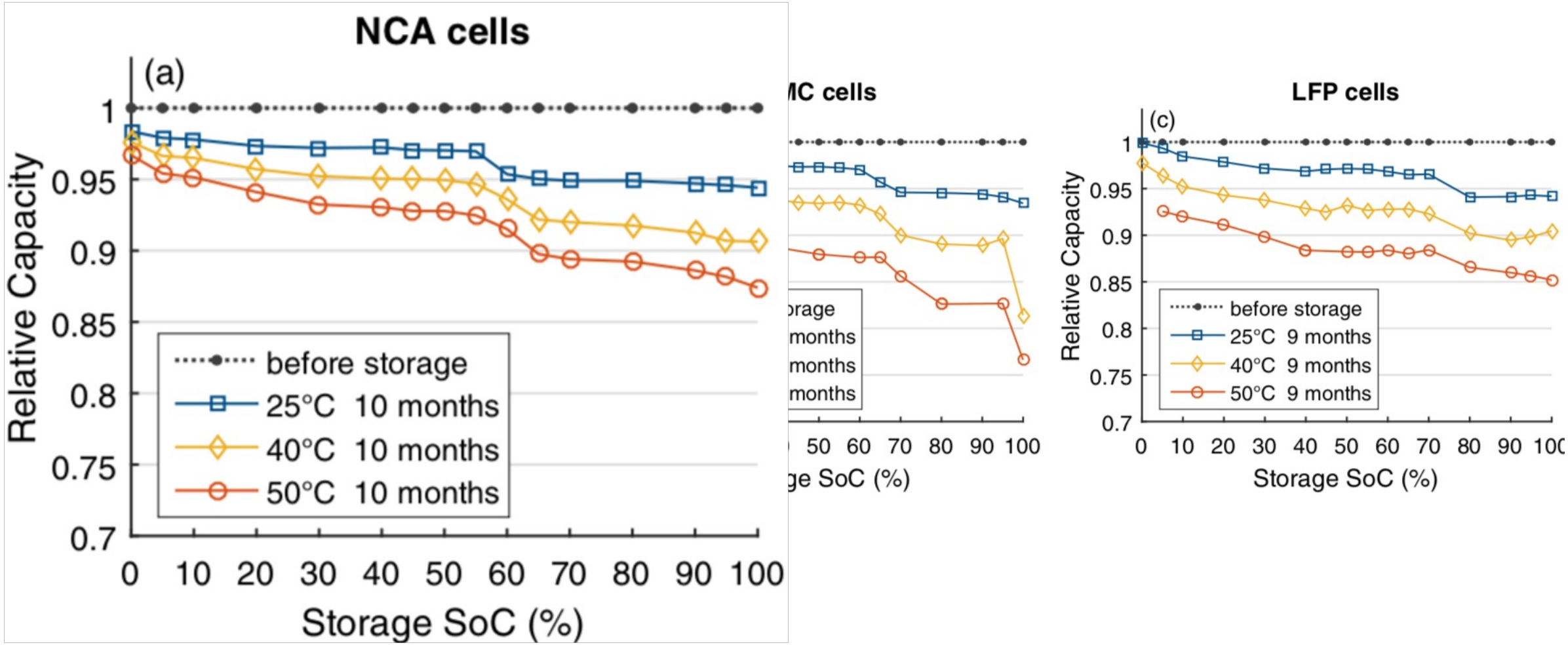

Fig. 2 in DOI: 10.1149/2.0411609jes [Keil 2016]

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Capacity loss: **Time**, SoC, Temperature, Chemistry

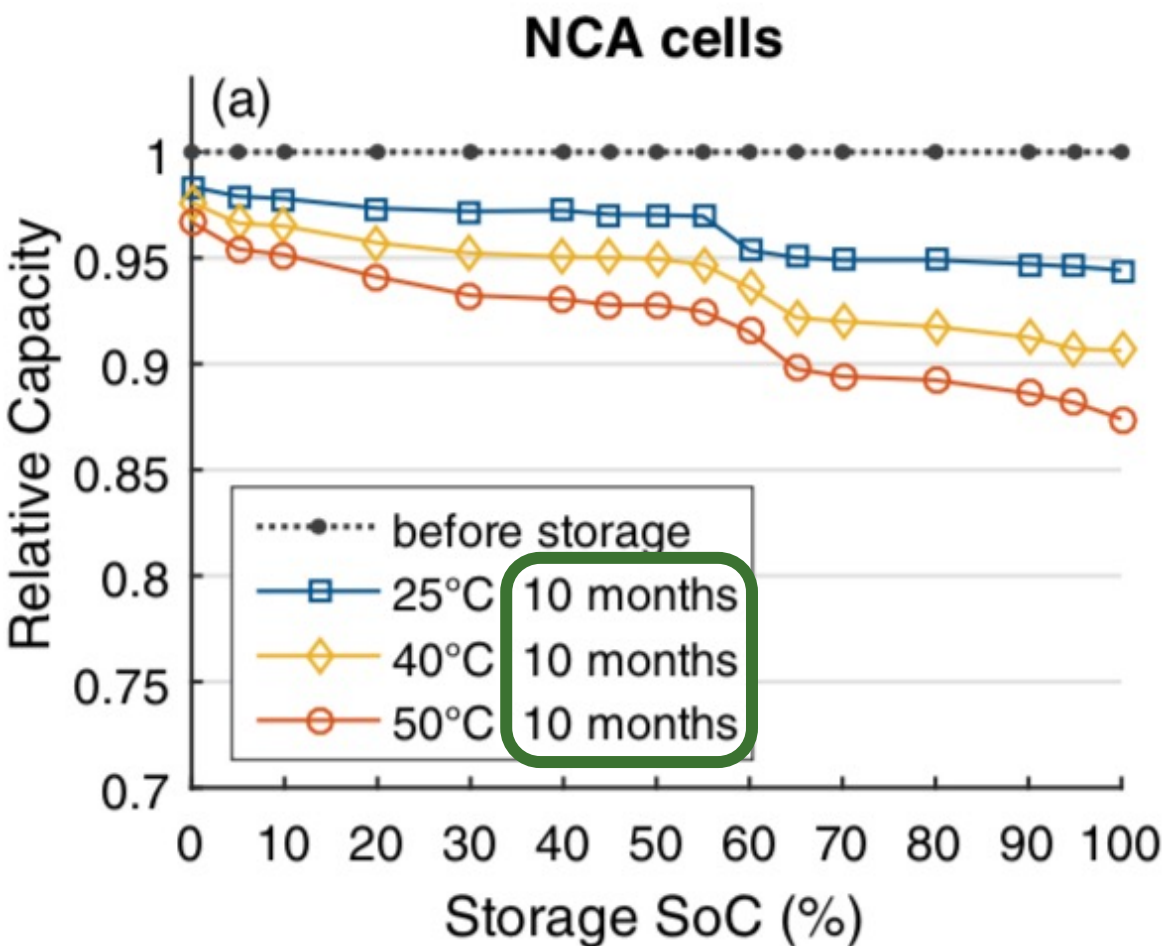

Fig. 2 in DOI: 10.1149/2.0411609jes [Keil 2016]

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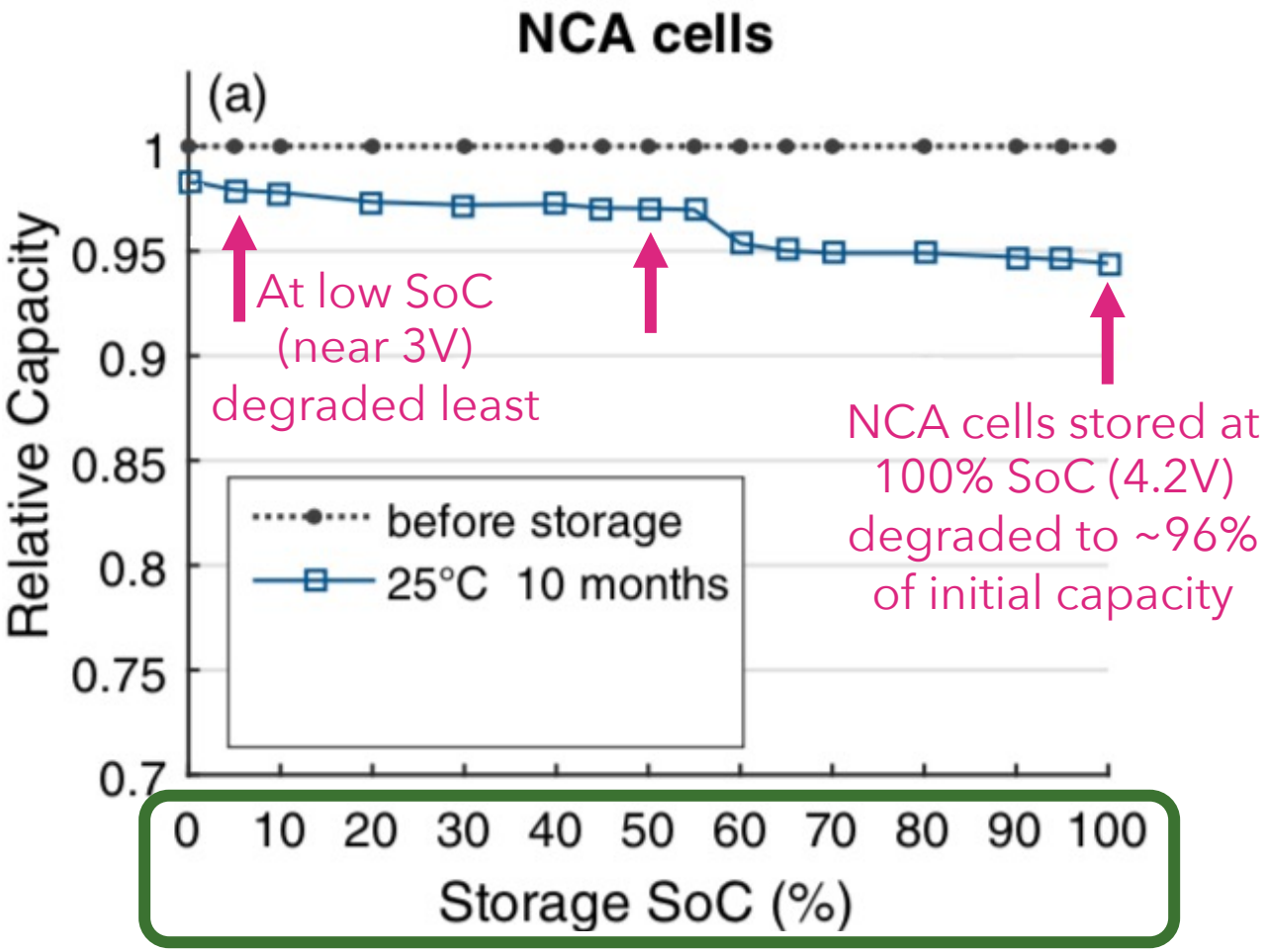
Capacity loss: Time, **SoC**, Temperature, Chemistry

Aging accelerates with state of charge.



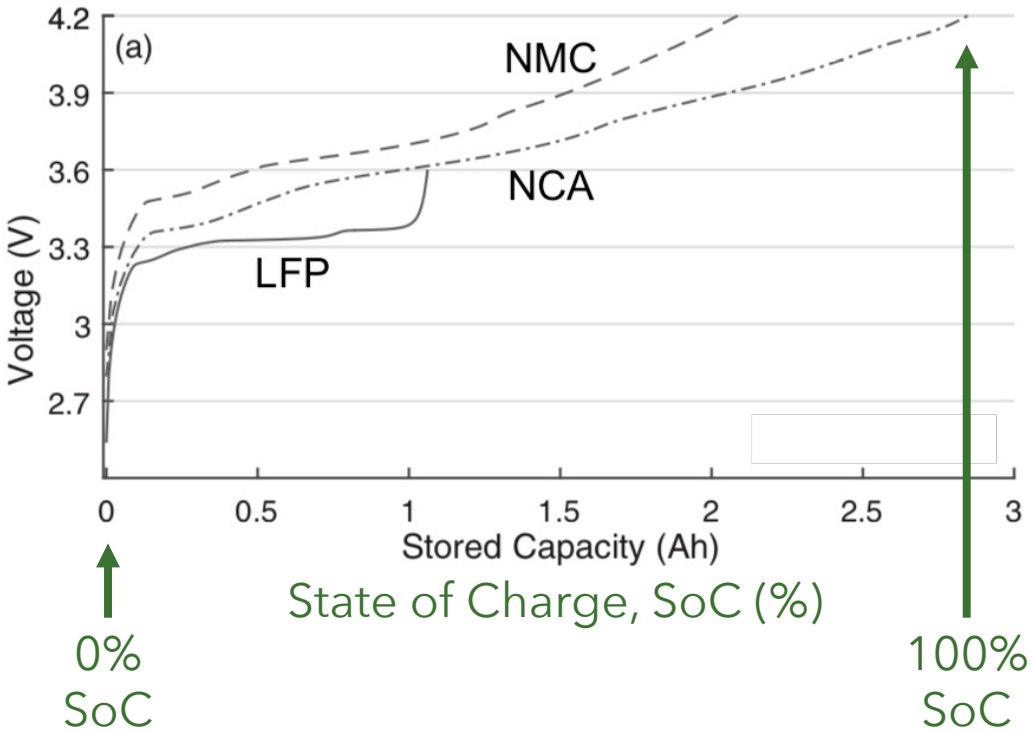


Fig. 2 in DOI: 10.1149/2.0411609jes [Keil 2016]

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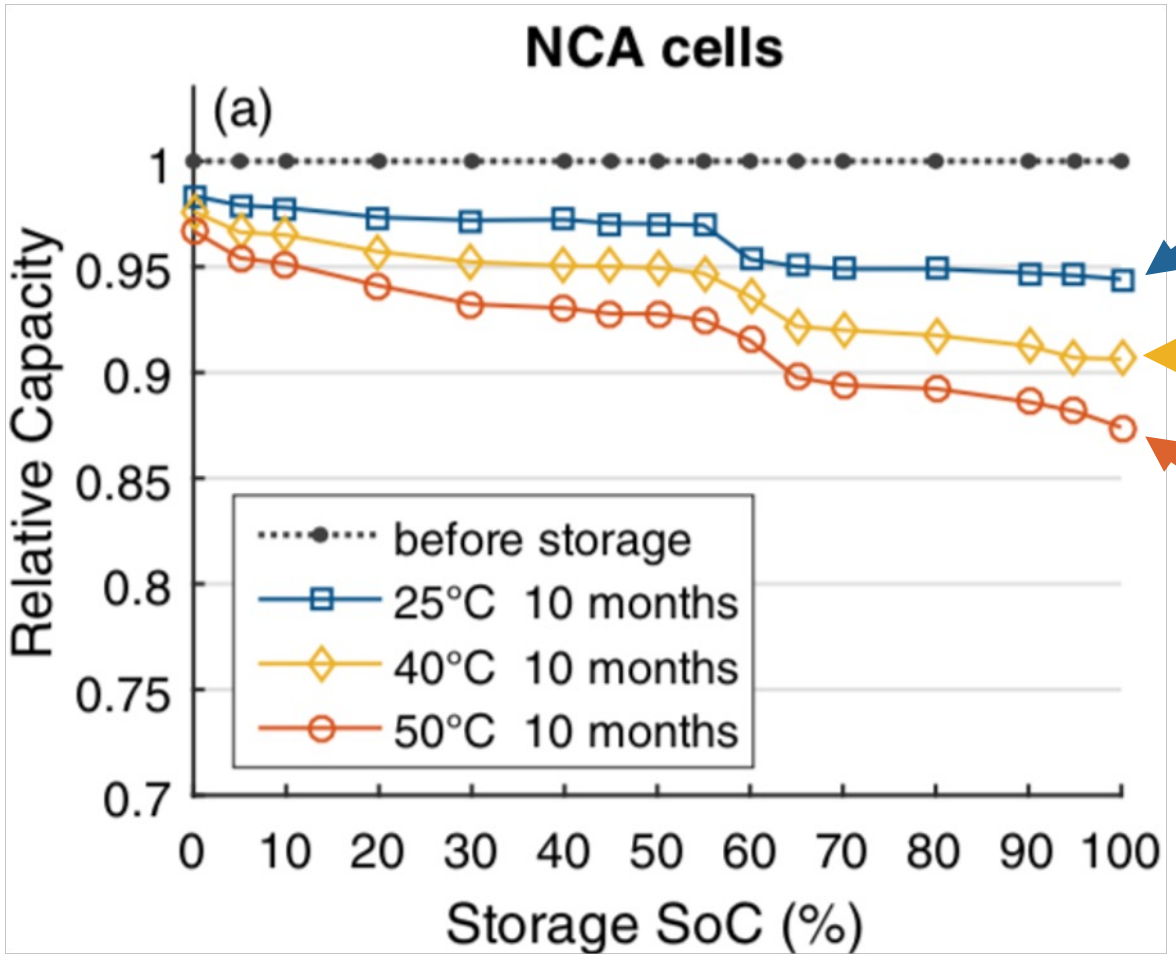
Capacity loss: Time, SoC, **Temperature**, Chemistry

Aging accelerates with temperature.



At 25°C, cells degraded to ~96% of initial capacity

At 40°C, cells degraded to ~91% of initial capacity

At 50°C, cells degraded to ~87% of initial capacity

Fig. 2 in DOI: 10.1149/2.0411609jes [Keil 2016]

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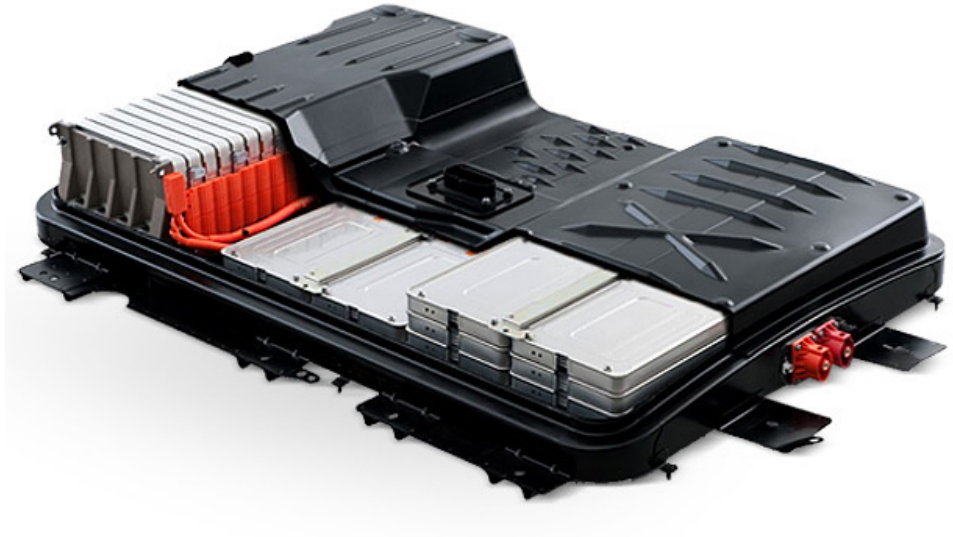
Capacity loss: Case study (Nissan Leaf in Arizona)

2011-2012 Nissan Leaf

- Reports of 1 of 12 bars (8.3%) capacity loss in 1st year
- Nissan initially claimed a dashboard instrumentation problem
- Leaf owners performed a controlled range test in Phoenix, confirming range loss.



What did Nissan's mechanical engineers do differently than other car makers?

Leaf: Air-cooled battery

Pause & Think

https://www.technologyreview.com/2012/09/19/85161/are-air-cooled-batteries-hurting-nissan-leaf-range/

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Capacity loss: Time, SoC, Temperature, **Chemistry**

Pause & Think

At 25C, all age similarly

NMC is most sensitive to high temperature

NMC is most sensitive to high SoC

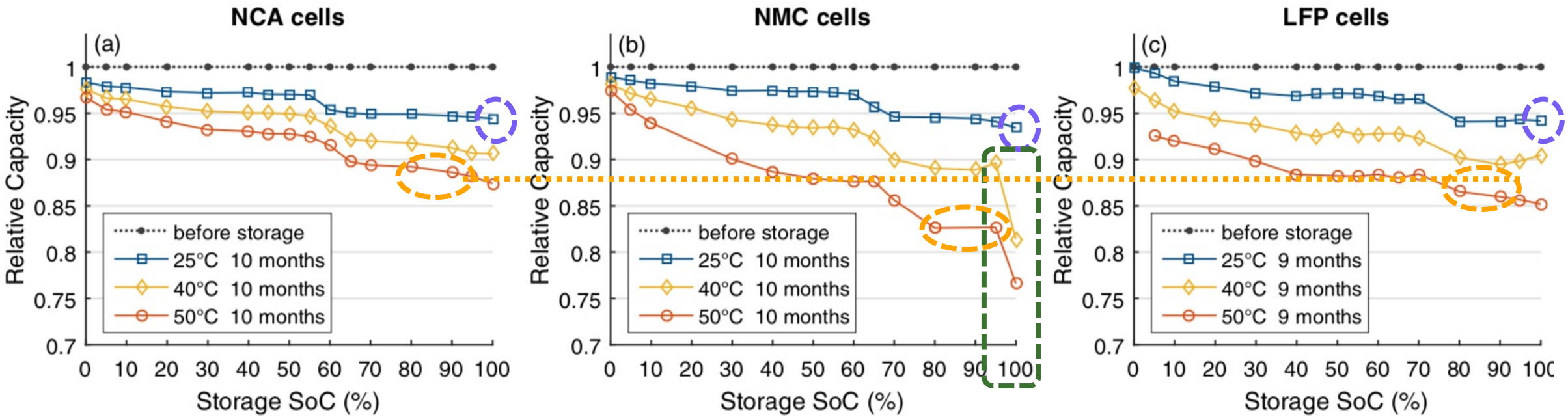

Fig. 2 in DOI: 10.1149/2.0411609jes [Keil 2016]

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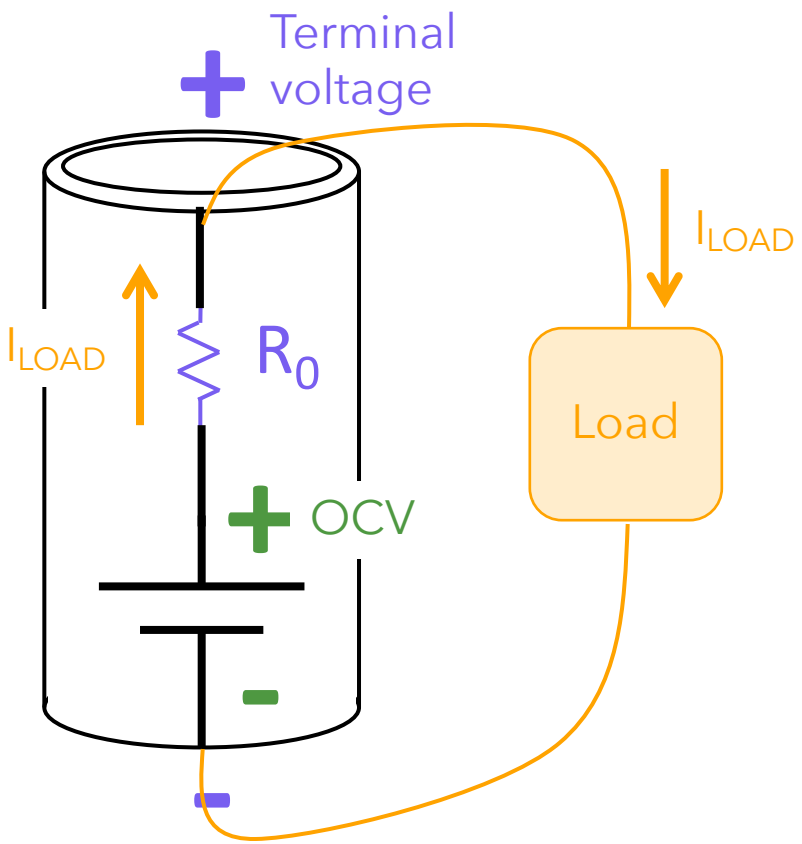
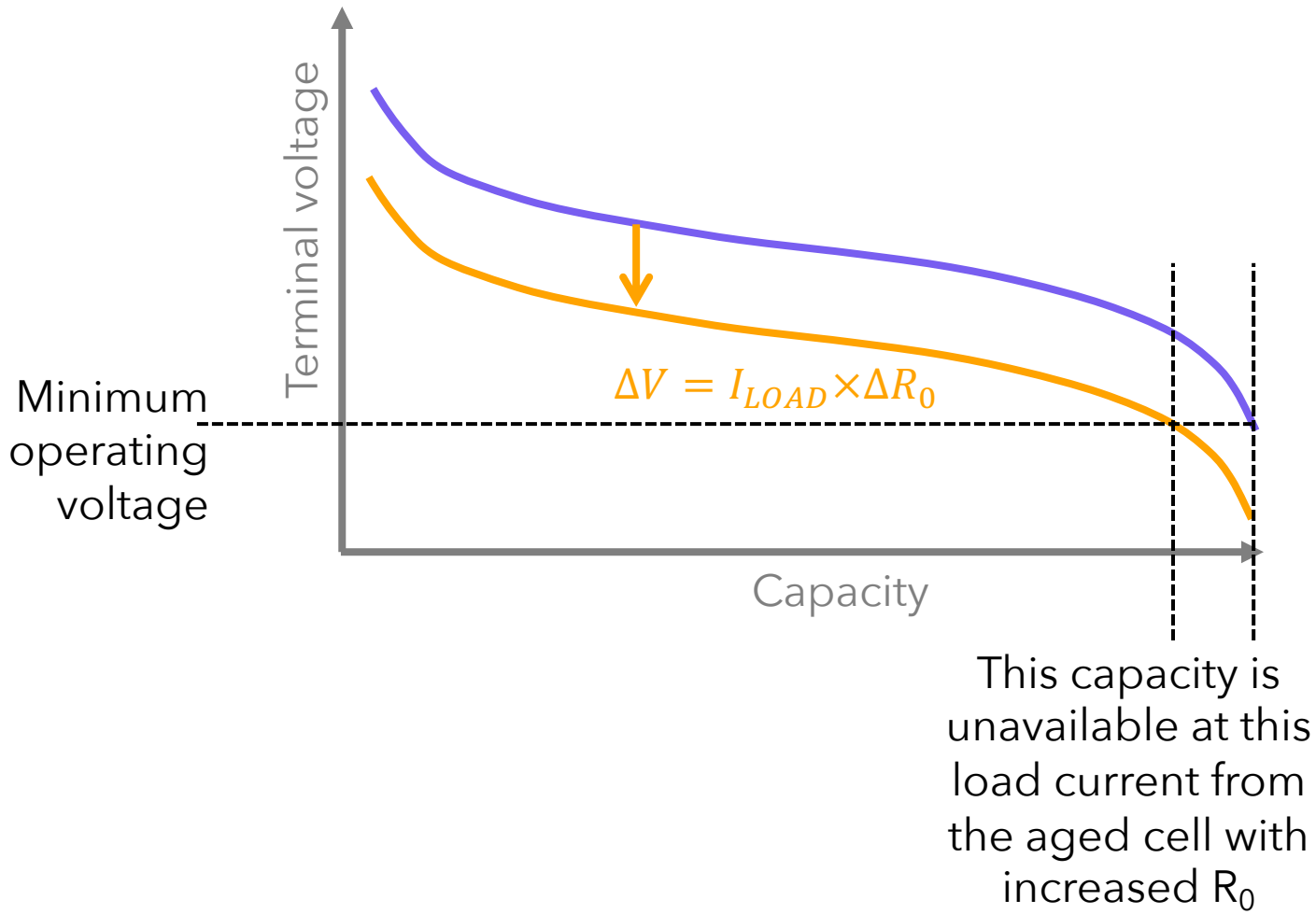
https://ofb.net/~ania/teaching/BMS-course/

Aging: Cell impedance

Increased I×R drop of an aged cell impairs access to all available capacity and/or reduces available power at low SOC.

Pause & Think





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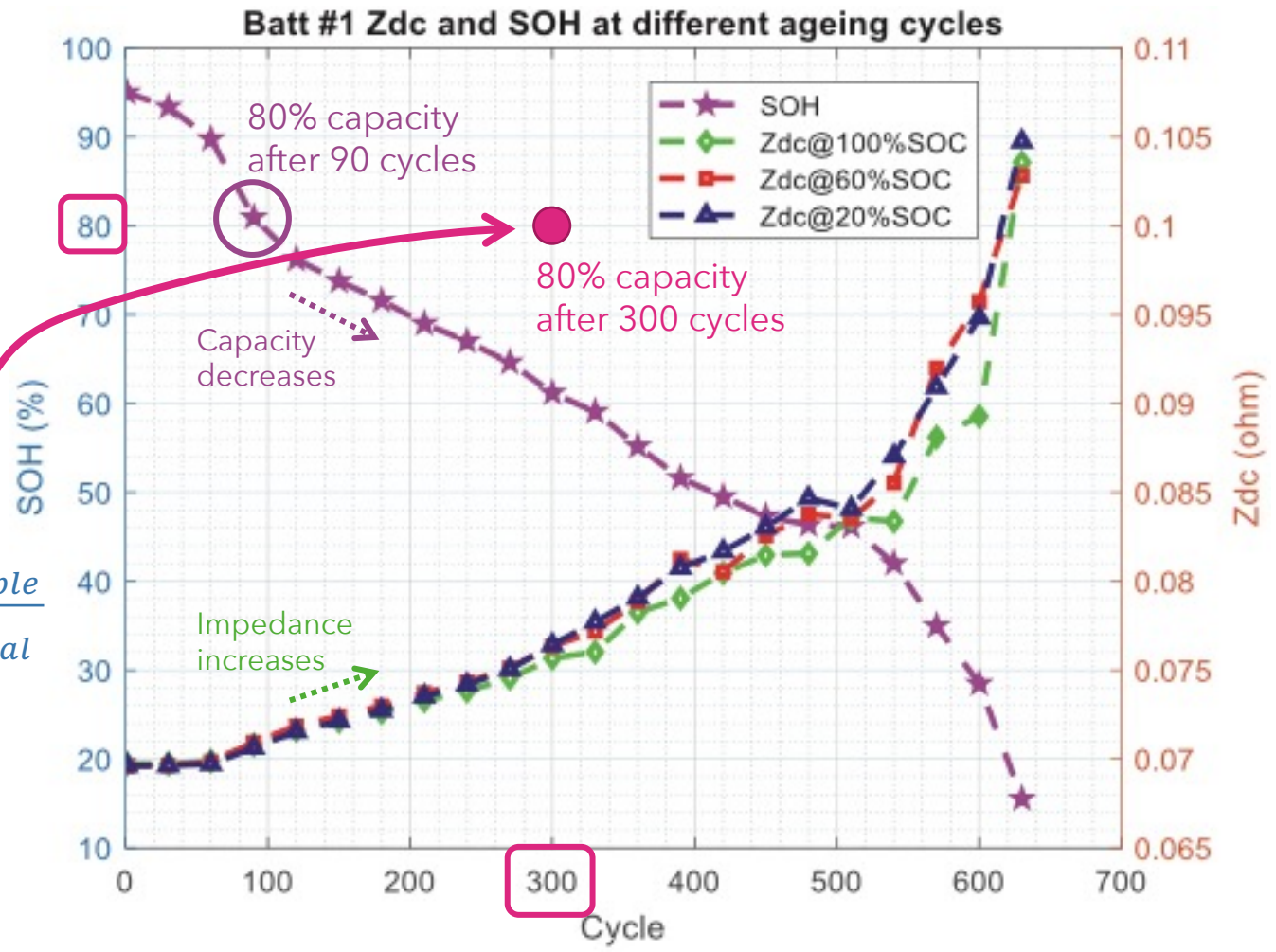
Aging: Capacity & impedance data

Measured aging of a 3.7V 2600mAh cylindrical lithium ion cell (Tenergy ICR 16,650-2600).

- 3.7V → NMC or similar chemistry
- Authors charged and discharged at 1C.
- Datasheet cycle life: 300th cycle at ≥80% of initial capacity (**0.5C**/1.0C at 25°C)

$$SOH = \frac{Q_{available}}{Q_{nominal}}$$



From Xia 2021, https://doi.org/10.1016/j.dib.2021.107727

Aging affects: Capacity & impedance

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Cell behaviors and models

A basic understanding of some battery cell behaviors, enough to optimize battery management architecture.

- Terminology
- OCV: Open Circuit Voltage
- Cell impedance
- Time-dependent responses
- Aging
- **Another learning resource**

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Another learning resource

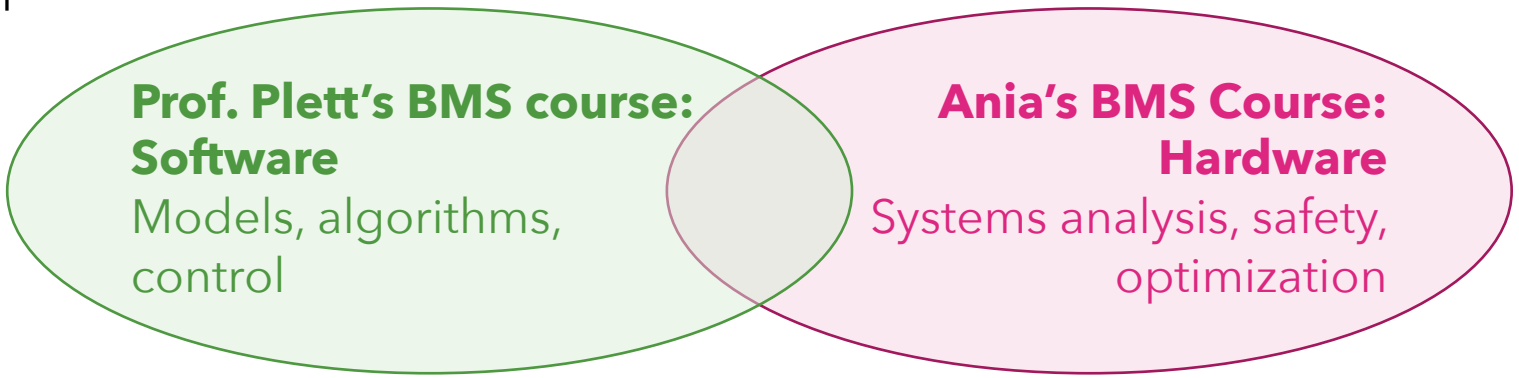
Professor Gregory Plett at University of Colorado in Colorado Springs also teaches a BMS course.

University class

- http://mocha-java.uccs.edu/ECE5720/ ← Cell model math
  - "Simulating Battery Packs"

Coursera

- https://www.coursera.org/instructor/gplett
  - Introduction to battery management systems
  - Equivalent circuit cell model simulation
  - Battery state of charge (SOC) estimation
  - Battery state of health (SOH) estimation
  - Battery pack balancing and power estimation



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"The End" by Gerd Altmann from Pixabay